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DESCRIPTION

📌 Goal

The goal of this project is to demonstrate how to create an **API** for an **e-commerce** shop that sells **virtual products** (such as eBooks) using a **microservices architecture**.

The implementation is based on:

- **Java**
- **Spring Boot 3**
- **Spring Cloud**

The project showcases a complete microservices ecosystem including business services, infrastructure components, observability tools, and external integrations.

🧩 Services Overview

1️⃣ Custom (Business) Services

These services implement the core business logic of the e-commerce platform:

- **product-service**
  Manages virtual products (e.g., eBooks).
- **order-service**
  Handles customer orders.
- **payment-service**
  Processes payments.
- **delivery-service**
  Manages delivery of virtual products.
- **notification-service**
  Sends email notifications to customers.

2️⃣ Architecture Services

These services support the microservices ecosystem and enable service coordination:

- **config-server**
  Centralized configuration management for all services.
- **discovery-service**
  Service registry that maintains a list of all running services in the ecosystem.

- **api-gateway**
  Single entry point from the Internet to the microservices ecosystem.
  Handles routing, filtering, and security at the edge.

3️⃣ Infrastructure Services

These services provide security, messaging, storage, and observability:

- **keycloak**
  Manages authentication and authorization.
- **kafka**
  Enables asynchronous communication between services.
- **kafka-ui**
  Provides a user interface for monitoring Kafka topics and messages.
- **postgres**
  SQL database used for storing order data.
- **mongo**
  NoSQL database used for storing product data.
- **grafana**
  Dashboard UI for distributed logging, monitoring, and performance analysis.
- **prometheus**
  Collects and measures performance metrics.
- **loki**
  Aggregates logs from all services.
- **tempo**
  Collects and tracks distributed traces.

4️⃣ External Services

These third-party services are integrated into the system:

- **stripe**
  External payment processing platform.
- **mailtrap**
  Email testing and delivery platform.

🖥️ Available Consoles

The **ecommerce-virtual** project provides the following user interfaces:

Service	Purpose	URL

Grafana (with Loki, Tempo, and Prometheus)	Distributed logging, monitoring & performance	http://localhost:3000
Kafka UI	Asynchronous communication monitoring	http://localhost:9000
Keycloak	Authentication & security management	http://localhost:7070
Stripe	Payment management	https://stripe.com
Mailtrap	Email testing	https://mailtrap.io

🏗️ Architecture Summary

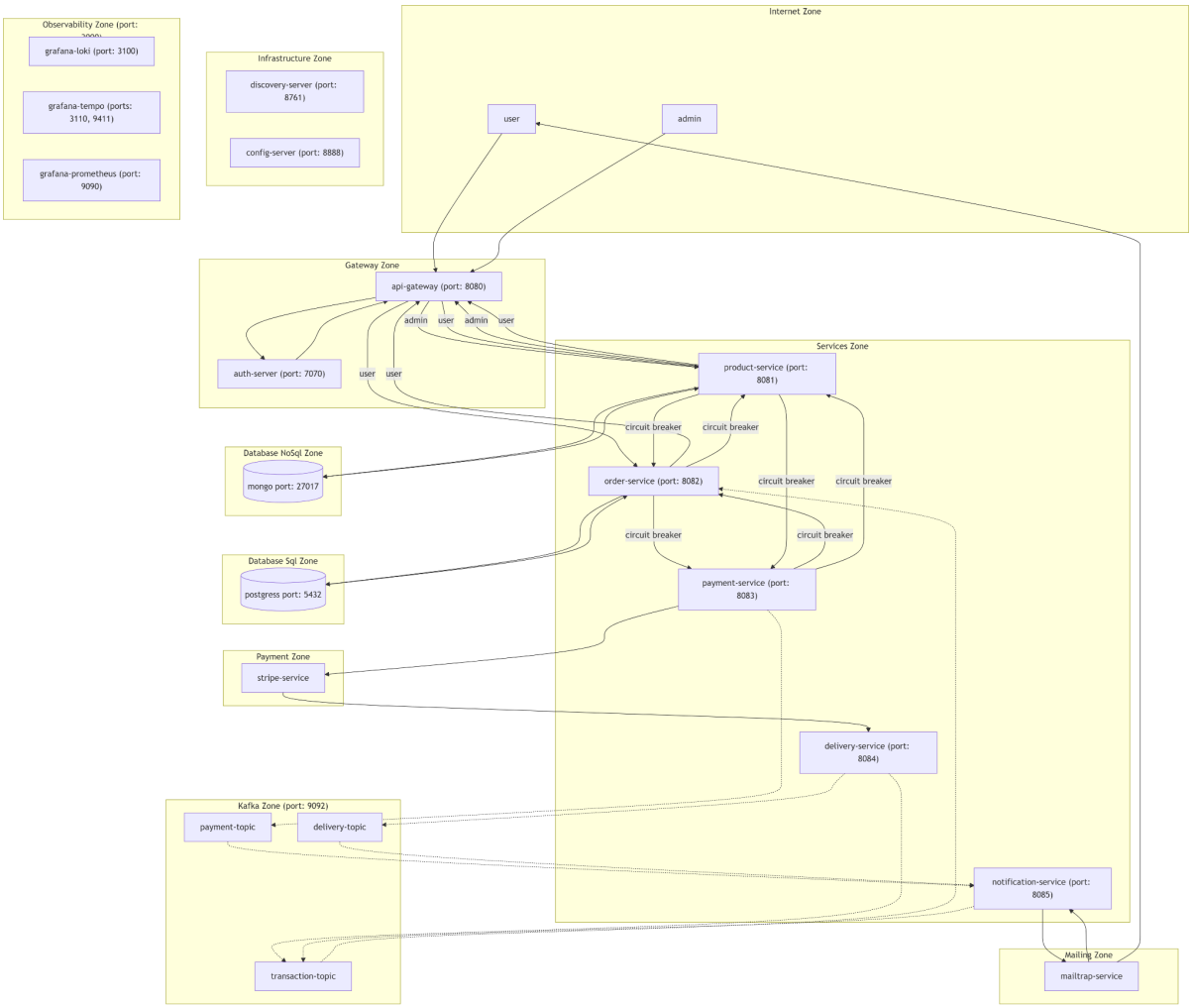
This project demonstrates:

- Microservices architecture
- API Gateway pattern
- Centralized configuration management
- Service discovery
- Event-driven communication using Kafka
- Secure authentication and authorization with Keycloak
- Observability using Prometheus, Grafana, Loki, and Tempo
- Integration with external services (Stripe, Mailtrap)
- Polyglot persistence (PostgreSQL + MongoDB)

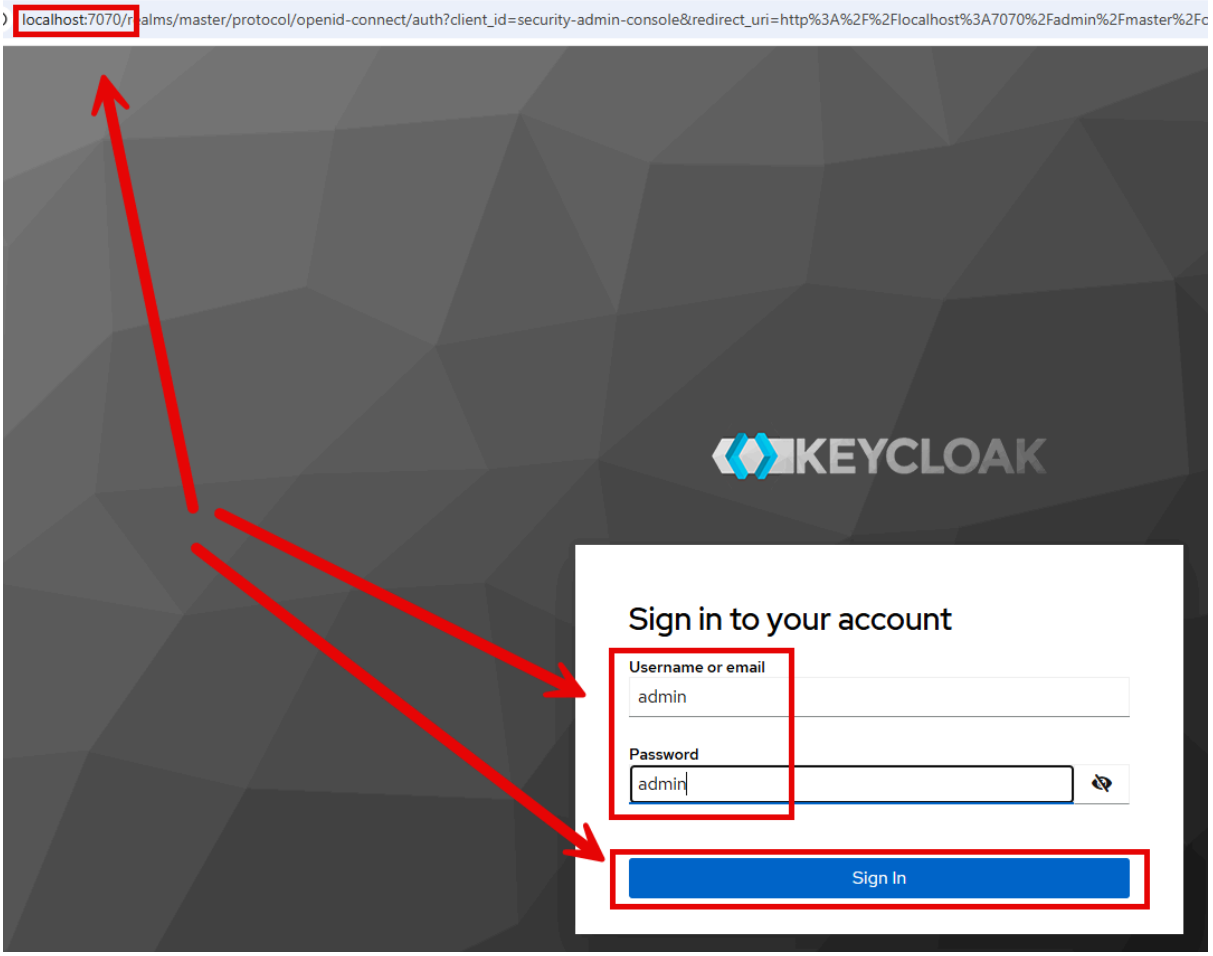
🎯 Project Benefits

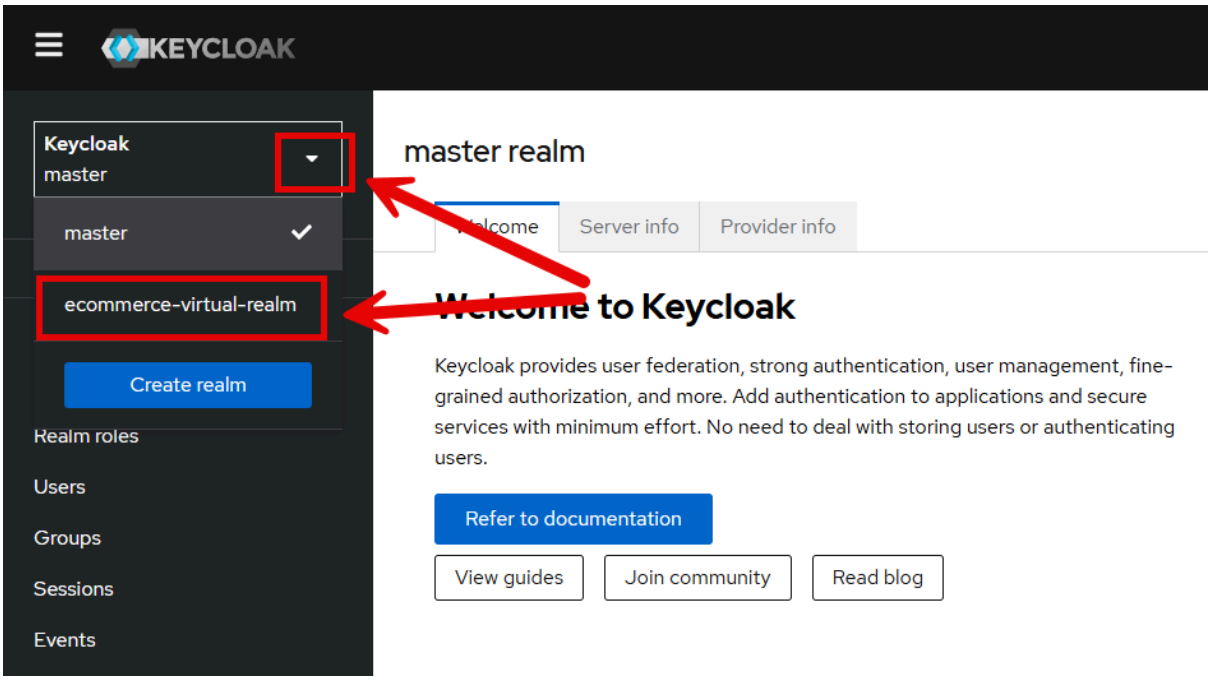
- Real-world microservices architecture example
- Full observability stack
- Secure authentication & authorization
- Event-driven communication
- Cloud-native ready design
- Clear separation of concerns between services

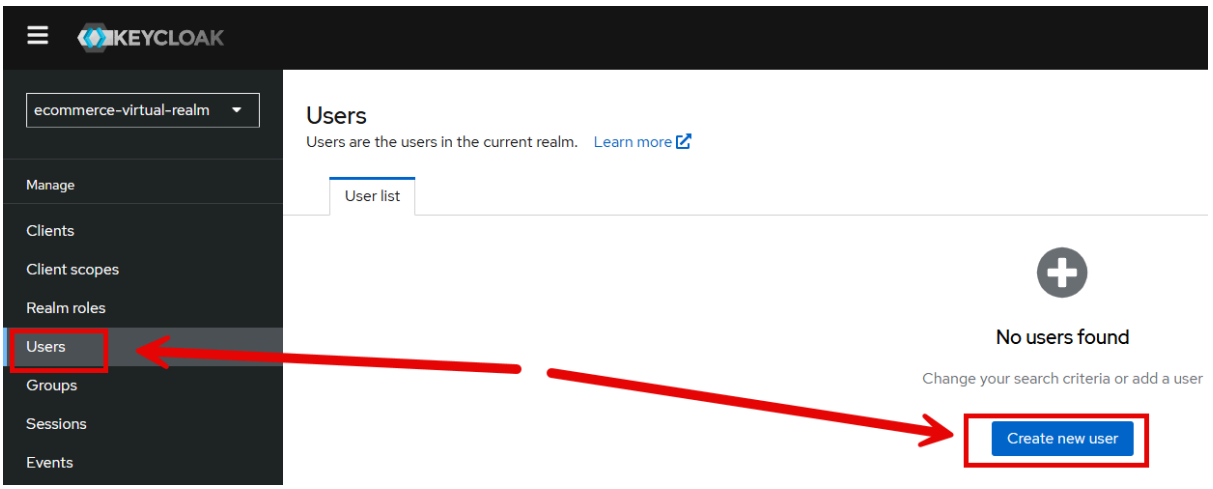
DIAGRAM

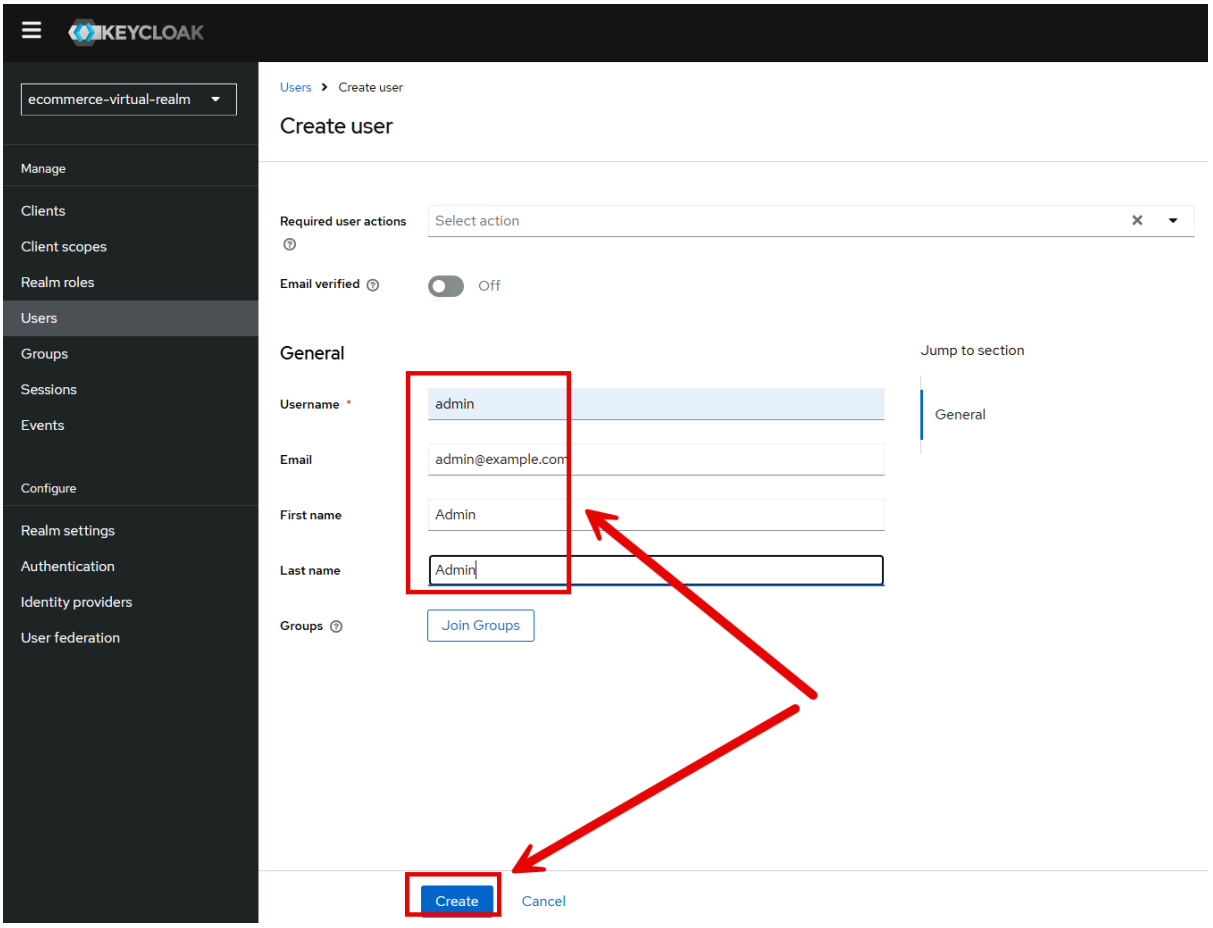

Observability Zone (port:
grafana-loki (port: 3100)
grafana-tempo (ports: 3110, 9411)
grafana-prometheus (port: 9090)
Infrastructure Zone
discovery-server (port: 8761)
config-server (port: 8888)
Internet Zone
user
admin
Gateway Zone
api-gateway (port: 8080)
admin
user
admin
user
auth-server (port: 7070)
user
user
Services Zone
product-service (port: 8081)
circuit breaker
circuit breaker
Database NoSql Zone
mongo port: 27017
order-service (port: 8082)
circuit breaker
circuit breaker
circuit breaker
circuit breaker
Database Sql Zone
postgress port: 5432
payment-service (port: 8083)
Payment Zone
stripe-service
delivery-service (port: 8084)
Kafka Zone (port: 9092)
payment-topic
delivery-topic
notification-service (port: 8085)
transaction-topic
Mailing Zone
mailtrap-service

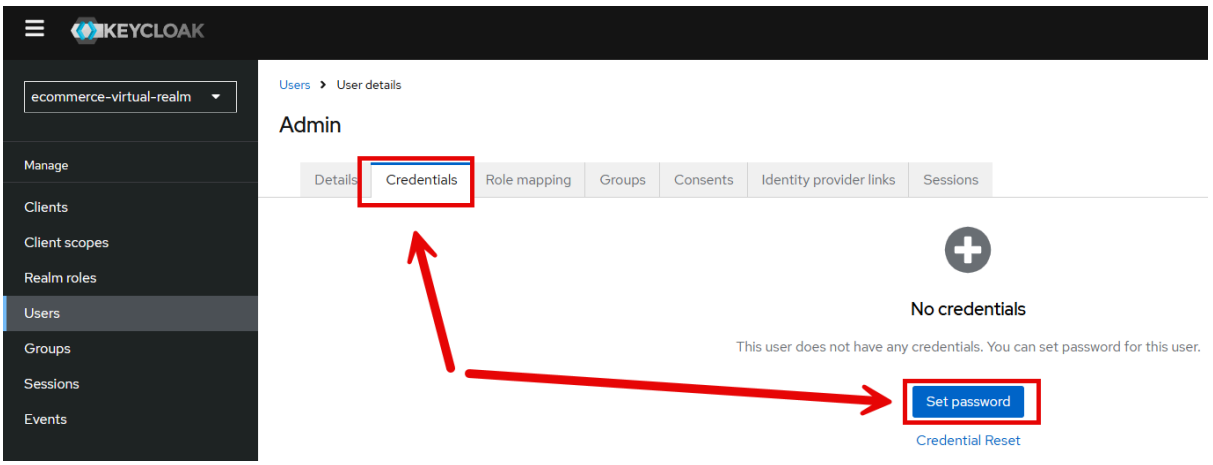
EXAMPLE POSTMAN

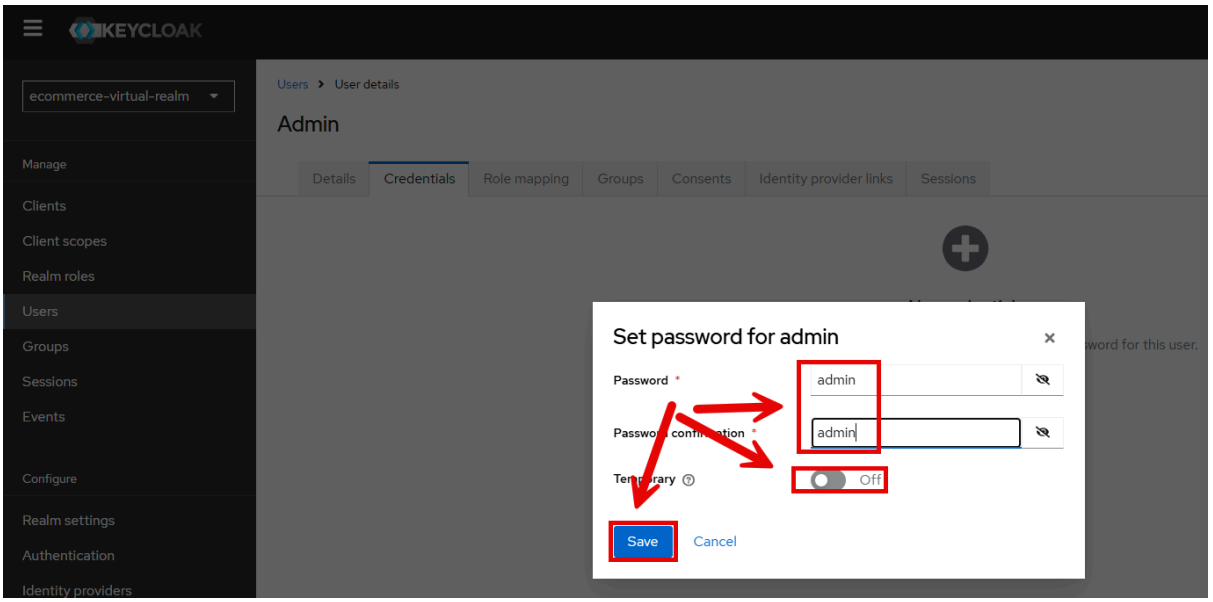


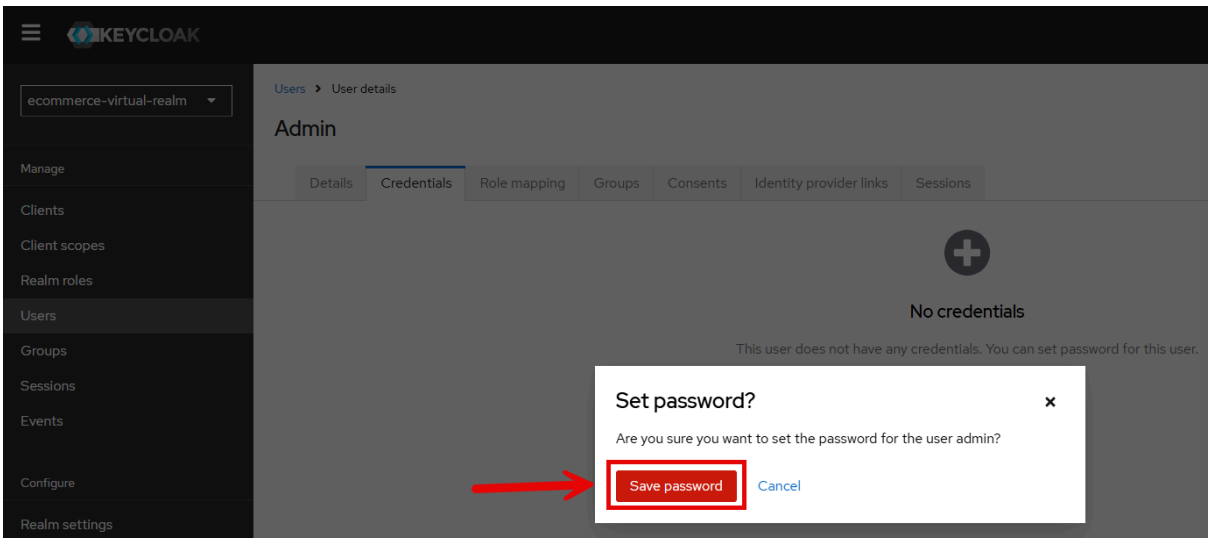



KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users
Users are the users in the current realm.
Learn more
User list
No users found
Change your search criteria or add a user
Create new user


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
User federation
Users > Create user
Create user
Required user actions
Select action
Email verified
Off
General
Jump to section
General
Username *
admin
Email
admin@example.com
First name
Admin
Last name
Admin
Groups
Join Groups
Create
Cancel


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
No credentials
This user does not have any credentials. You can set password for this user.
Set password
Credential Reset


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
Set password for admin
Password *
admin
admin
Off
Save
Cancel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Realm roles
Users
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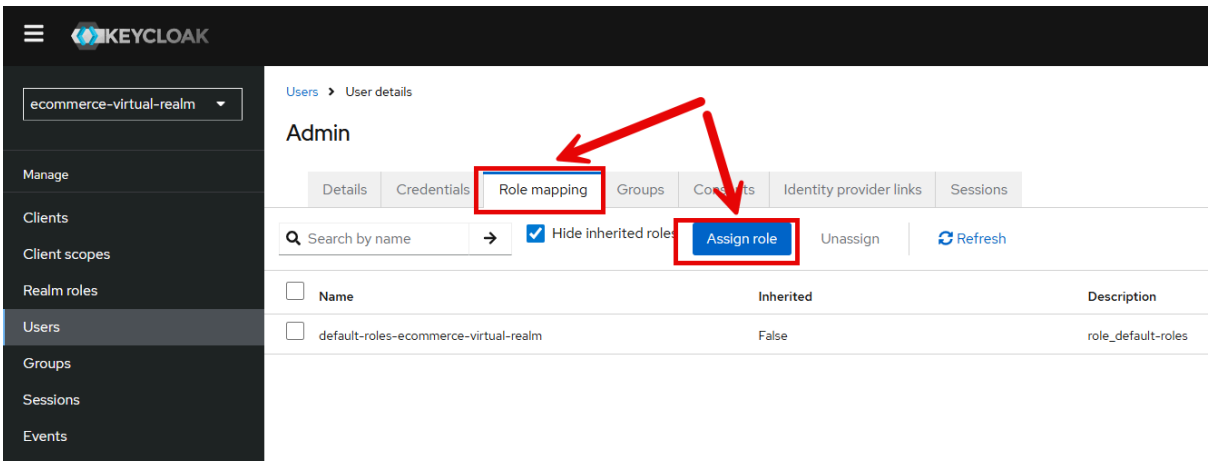
Configure
Realm settings
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
No credentials
This user does not have any credentials. You can set password for this user.
Set password?
Are you sure you want to set the password for the user admin?
Save password
Cancel


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Identity provider links
Sessions
Search by name
Hide inherited roles
Assign role
Unassign
Refresh
Name
Inherited
Description
default-roles-ecommerce-virtual-realm
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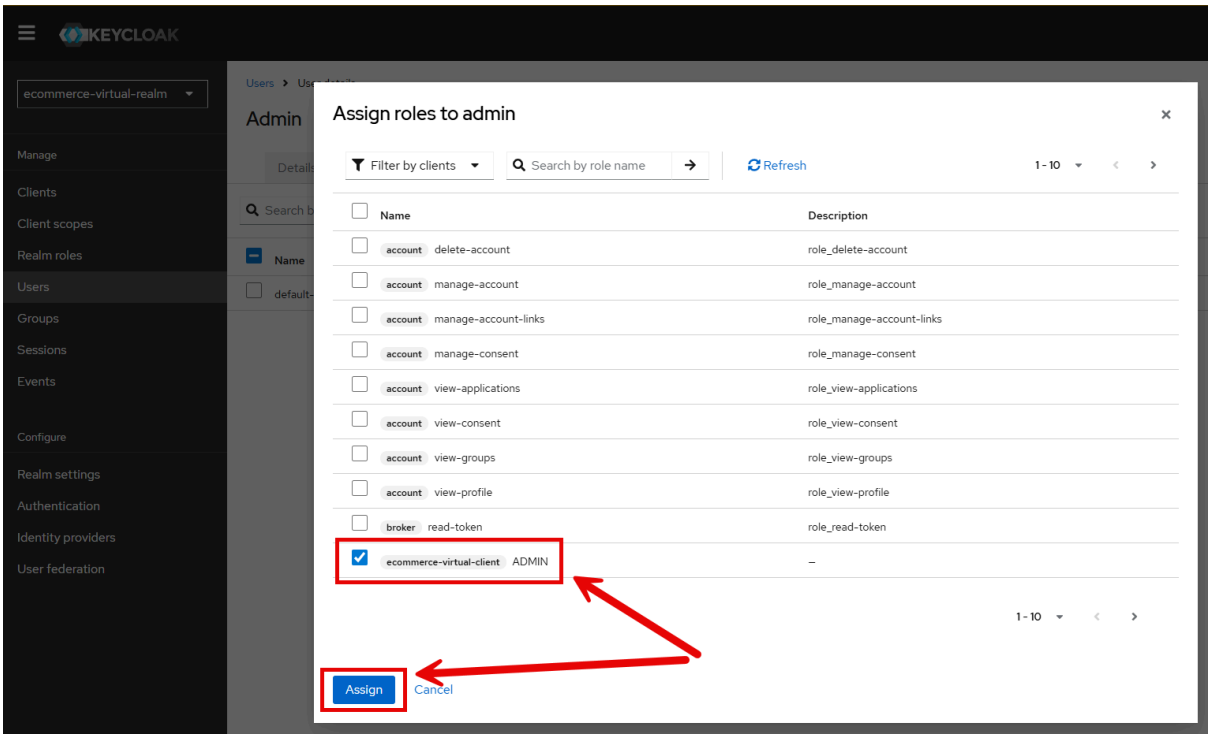
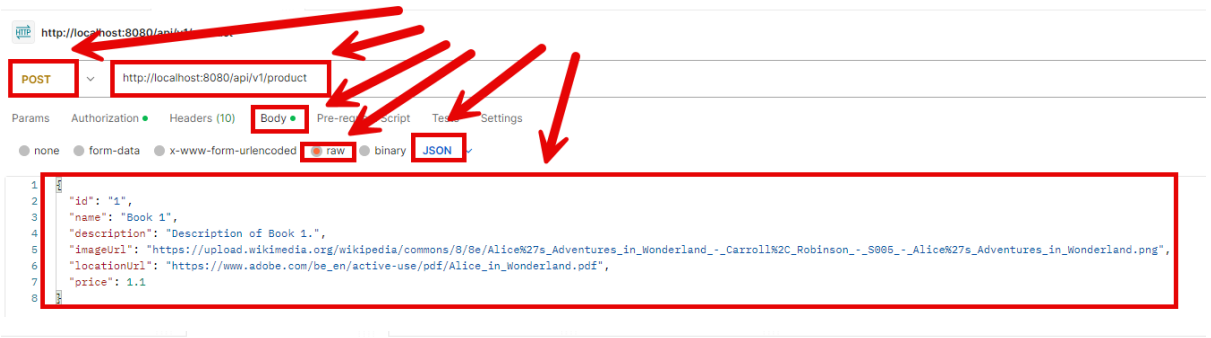
role_default-roles


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
User federation
Admin
Assign roles to admin
Filter by clients
Search by role name
Refresh
1 - 10
Name
Description
account delete-account
role_delete-account
account manage-account
role_manage-account
account manage-account-links
role_manage-account-links
account manage-consent
role_manage-consent
account view-applications
role_view-applications
account view-consent
role_view-consent
account view-groups
role_view-groups
account view-profile
role_view-profile
broker read-token
role_read-token
ecommerce-virtual-client ADMIN
–
Assign
Cancel


http://localhost:8080/api/v1/product
POST
http://localhost:8080/api/v1/product
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
none
form-data
x-www-form-urlencoded
raw
binary
JSON
{
"id": "1",
"name": "Book 1",
"description": "Description of Book 1.",
"imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
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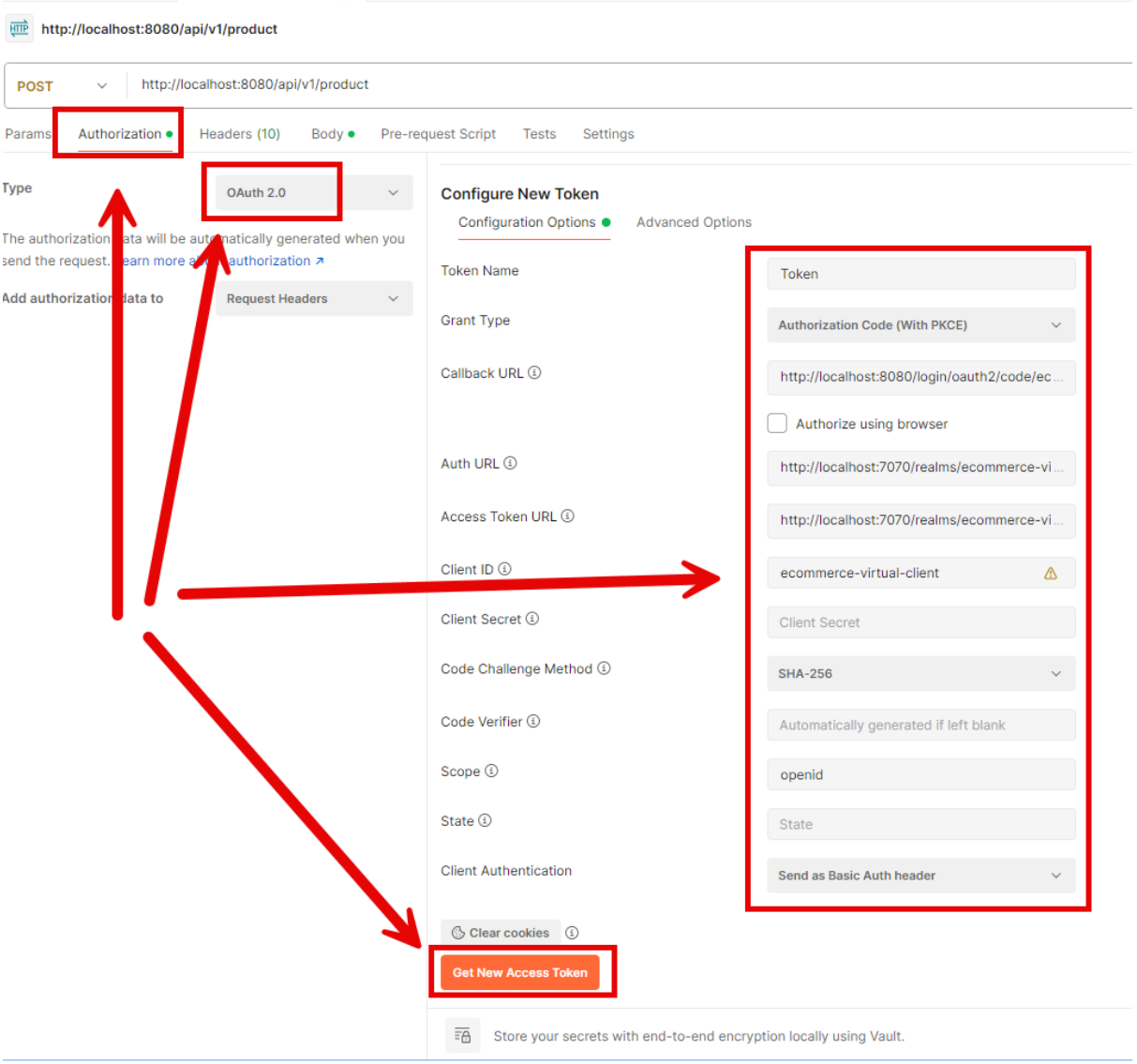
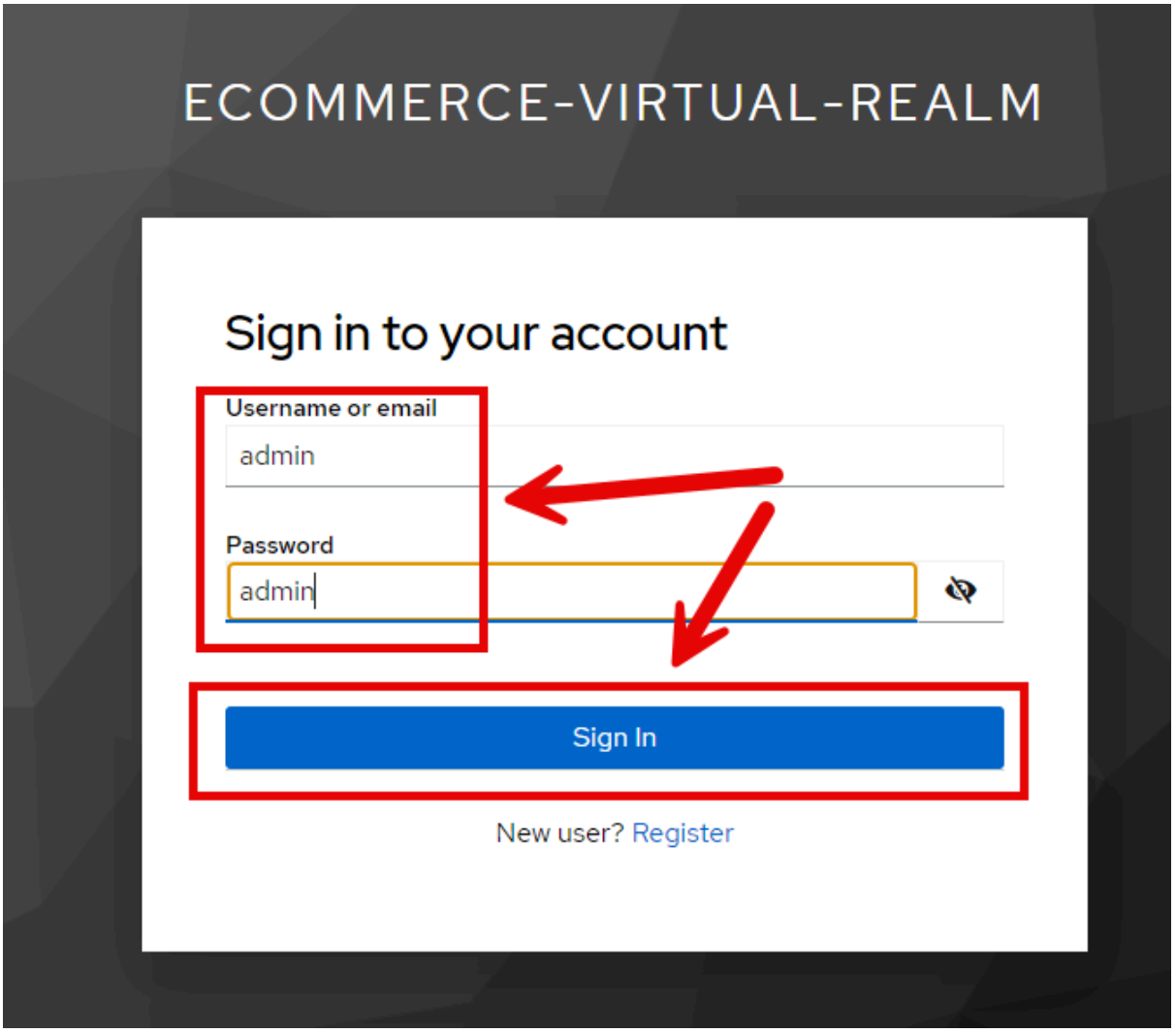
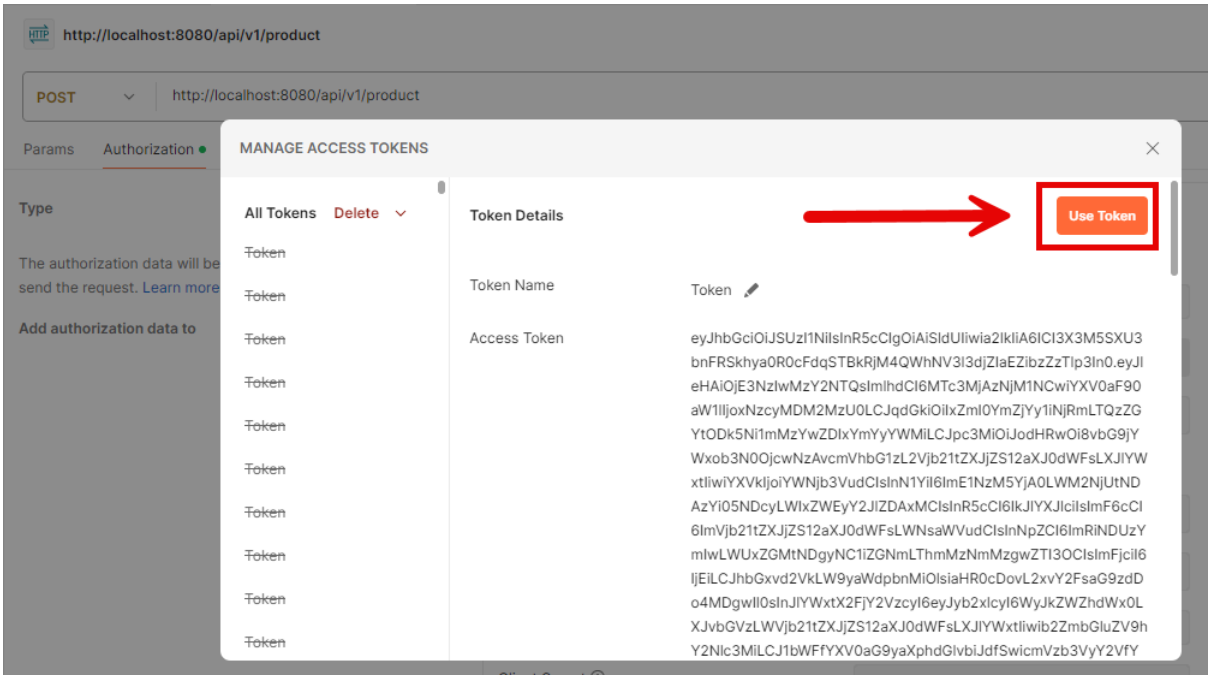
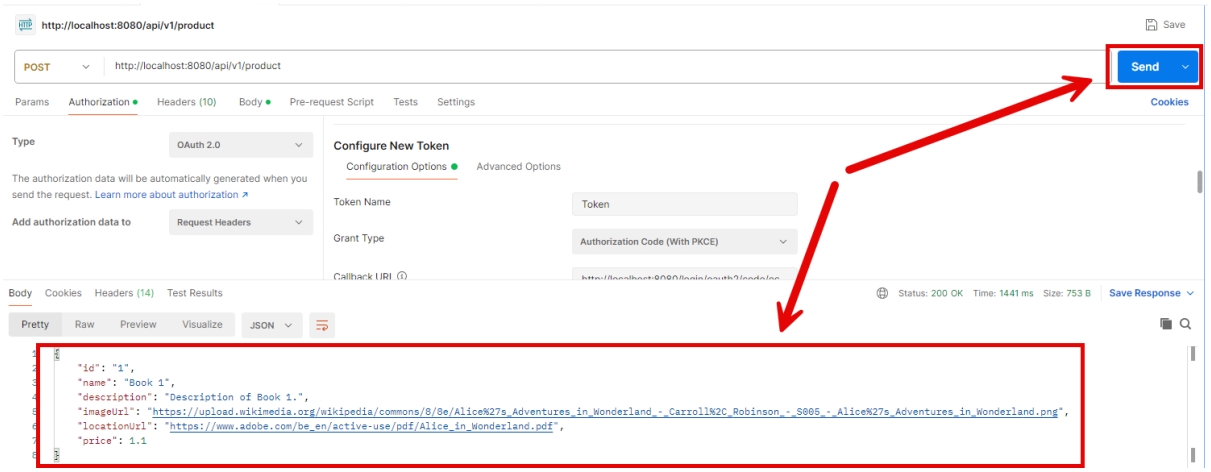
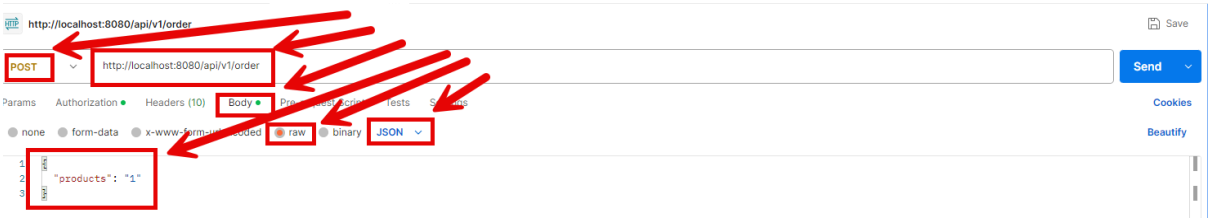
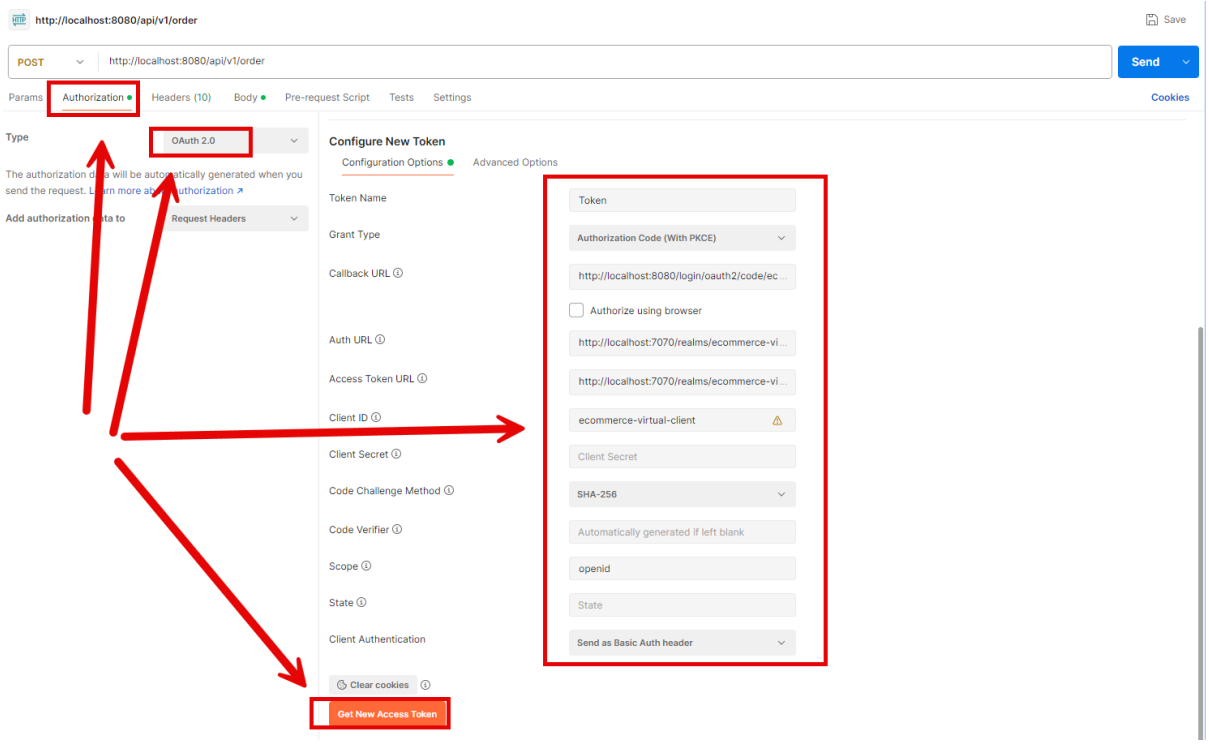

http://localhost:8080/api/v1/product
POST
http://localhost:8080/api/v1/product
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
Type
OAuth 2.0
The authorization data will be automatically generated when you send the request. Learn more about authorization
Add authorization data to
Request Headers
Configure New Token
Configuration Options
Advanced Options
Token Name
Token
Grant Type
Authorization Code (With PKCE)
Callback URL
http://localhost:8080/login/oauth2/code/ec...
Authorize using browser
Auth URL
http://localhost:7070/realms/ecommerce-vi...
Access Token URL
http://localhost:7070/realms/ecommerce-vi...
Client ID
ecommerce-virtual-client
Client Secret
Client Secret
Code Challenge Method
SHA-256
Code Verifier
Automatically generated if left blank
Scope
openid
State
State
Client Authentication
Send as Basic Auth header
Clear cookies
Get New Access Token
Store your secrets with end-to-end encryption locally using Vault.

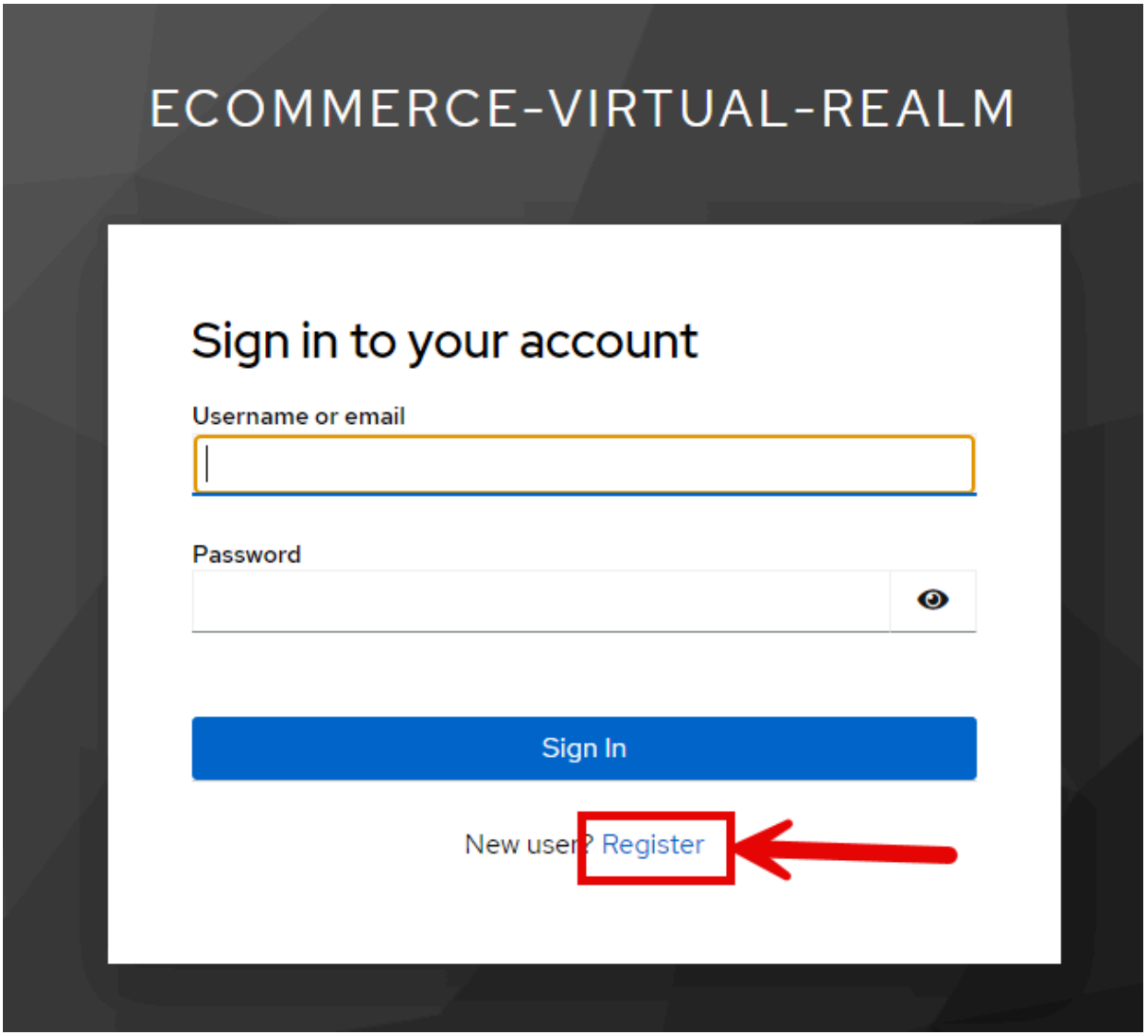

ECOMMERCE-VIRTUAL-REALM
Sign in to your account
Username or email
admin
Password
admin
Sign In
New user? Register


http://localhost:8080/api/v1/product
POST
http://localhost:8080/api/v1/product
Params
Authorization
MANAGE ACCESS TOKENS
Type
All Tokens
Delete
Token Details
Use Token
Token Name
Token
Access Token
Add authorization data to


http://localhost:8080/api/v1/product
Save
POST
http://localhost:8080/api/v1/product
Send
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
Cookies
Type
OAuth 2.0
Configure New Token
Configuration Options
Advanced Options
Token Name
Token
Grant Type
Authorization Code (With PKCE)
Add authorization data to
Request Headers
Body
Cookies
Headers (14)
Test Results
Status: 200 OK
Time: 1441 ms
Size: 753 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
"id": "1",
"name": "Book 1",
"description": "Description of Book 1.",
"imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
"locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
"price": 1.1


http://localhost:8080/api/v1/order
Save
POST
http://localhost:8080/api/v1/order
Send
Params
Authorization
Headers (10)
Body
Tests
Cookies
none
form-data
raw
binary
JSON
Beautify
"products": "1"


http://localhost:8080/api/v1/order
Save
POST
http://localhost:8080/api/v1/order
Send
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
Cookies
Type
OAuth 2.0
Add authorization data to
Request Headers
Configure New Token
Configuration Options
Advanced Options
Token Name
Token
Grant Type
Authorization Code (With PKCE)
Callback URL
http://localhost:8080/login/oauth2/code/ec...
Authorize using browser
Auth URL
http://localhost:7070/realms/ecommerce-vi...
Access Token URL
http://localhost:7070/realms/ecommerce-vi...
Client ID
ecommerce-virtual-client
Client Secret
Client Secret
Code Challenge Method
SHA-256
Code Verifier
Automatically generated if left blank
Scope
openid
State
State
Client Authentication
Send as Basic Auth header
Clear cookies
Get New Access Token


ECOMMERCE-VIRTUAL-REALM
Sign in to your account
Username or email
Password
Sign In
New user? Register

ECOMMERCE-VIRTUAL-REALM

Register

* Required fields

Username *

user

Password *

user

Confirm password *

user

Email *

user@example.com

First name *

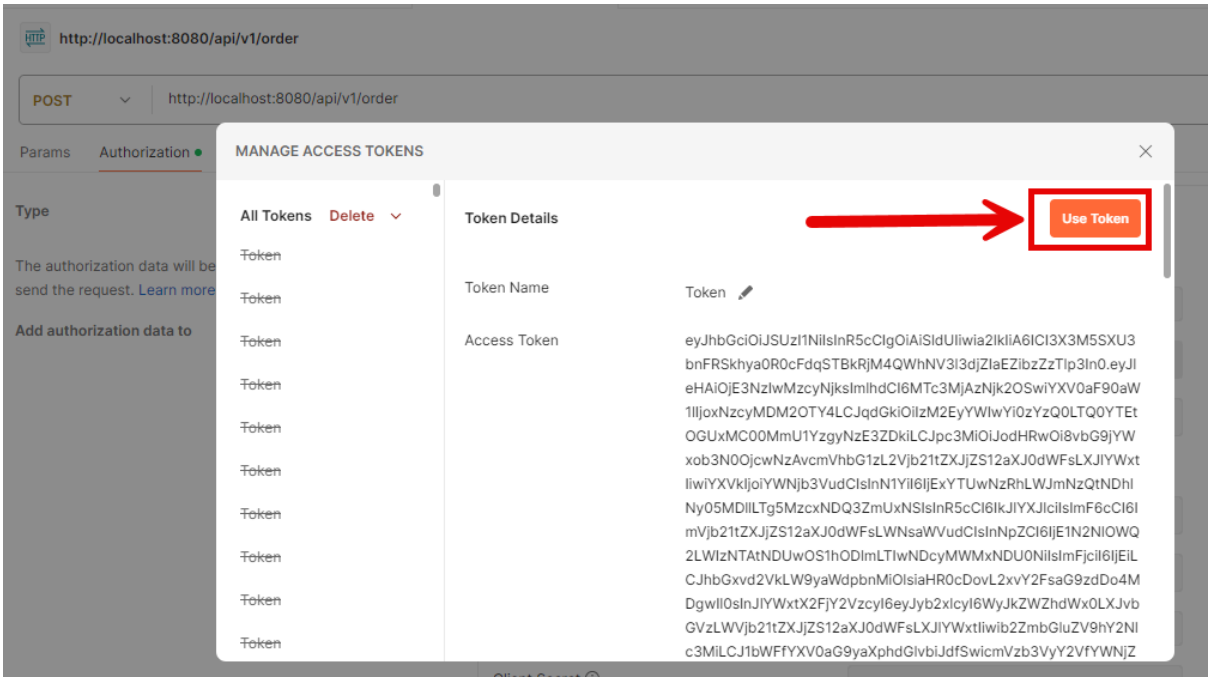
User

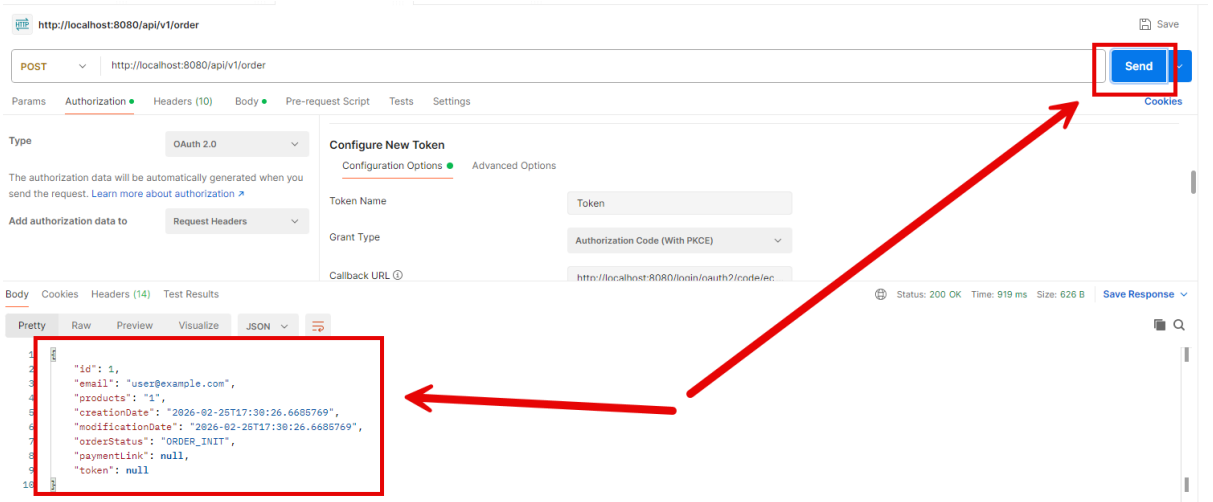
Last name *

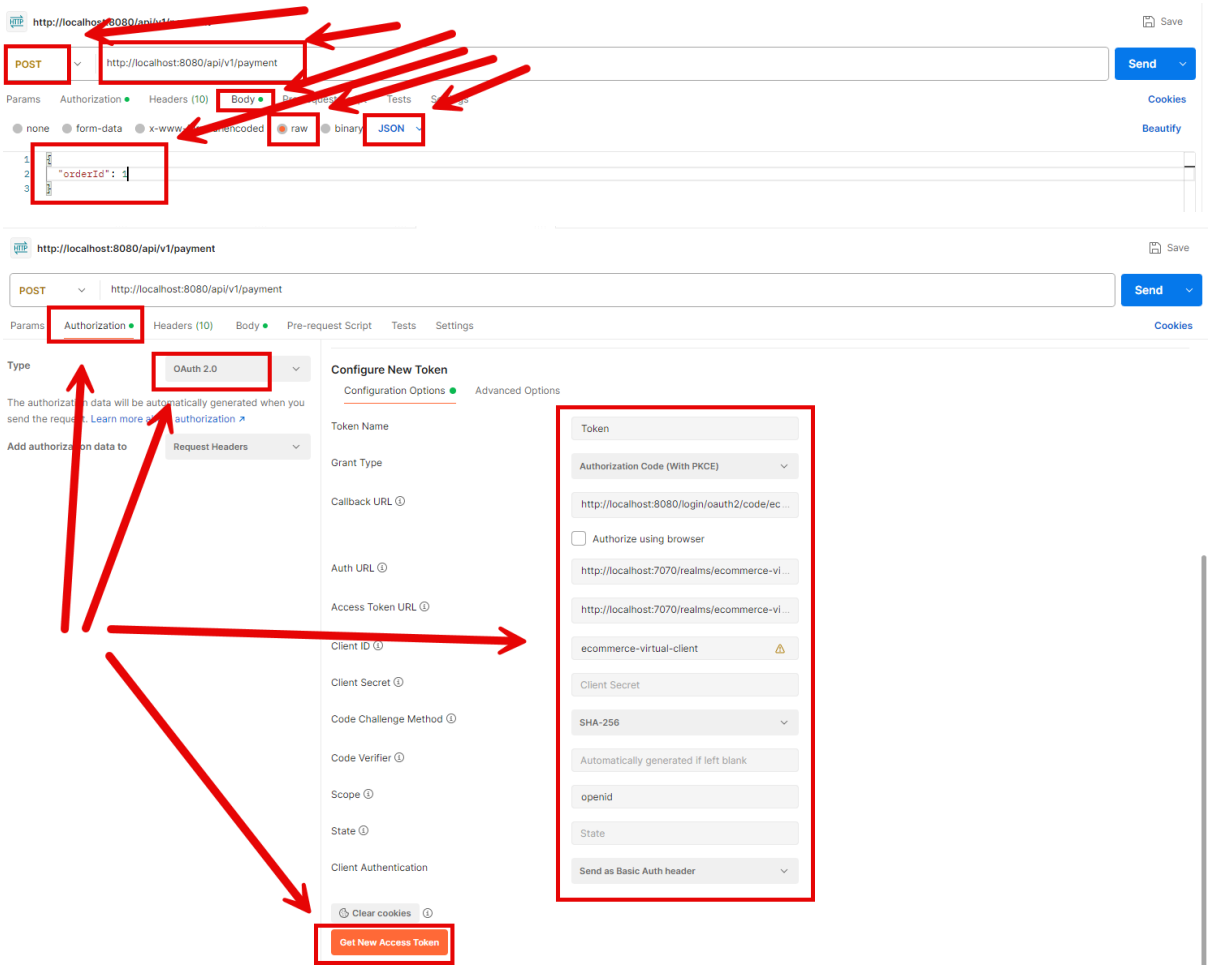
User

Register

« Back to Login


http://localhost:8080/api/v1/order
POST
http://localhost:8080/api/v1/order
Params
Authorization
MANAGE ACCESS TOKENS
Type
All Tokens
Delete
Token Details
Use Token
Token Name
Token
Access Token
Add authorization data to


http://localhost:8080/api/v1/order
Save
POST
http://localhost:8080/api/v1/order
Send
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
Cookies
Type
OAuth 2.0
Configure New Token
Configuration Options
Advanced Options
The authorization data will be automatically generated when you send the request. Learn more about authorization
Add authorization data to
Request Headers
Token Name
Token
Grant Type
Authorization Code (With PKCE)
Callback URL
Body
Cookies
Headers (14)
Test Results
Status: 200 OK
Time: 919 ms
Size: 626 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
"id": 1,
"email": "user@example.com",
"products": "1",
"creationDate": "2026-02-25T17:30:26.6685769",
"modificationDate": "2026-02-25T17:30:26.6685769",
"orderStatus": "ORDER_INIT",
"paymentLink": null,
"token": null


http://localhost:8080/api/v1/payment
Save
POST
http://localhost:8080/api/v1/payment
Send
Params
Authorization
Headers (10)
Body
Tests
Cookies
none
form-data
raw
binary
JSON
Beautify
"orderId": 5
http://localhost:8080/api/v1/payment
Save
POST
http://localhost:8080/api/v1/payment
Send
Params
Authorization
Headers (10)
Body
Pre-request Script
Tests
Settings
Cookies
Type
OAuth 2.0
Add authorization data to
Request Headers
Configure New Token
Configuration Options
Advanced Options
Token Name
Token
Grant Type
Authorization Code (With PKCE)
Callback URL
http://localhost:8080/login/oauth2/code/ec ...
Authorize using browser
Auth URL
http://localhost:7070/realms/ecommerce-vi ...
Access Token URL
http://localhost:7070/realms/ecommerce-vi ...
Client ID
ecommerce-virtual-client
Client Secret
Client Secret
Code Challenge Method
SHA-256
Code Verifier
Automatically generated if left blank
Scope
openid
State
State
Client Authentication
Send as Basic Auth header
Clear cookies
Get New Access Token

http://localhost:8080/api/v1/payment

POST http://localhost:8080/api/v1/payment

Params Authorization

Type

The authorization data will be sent the request. Learn more

Add authorization data to

MANAGE ACCESS TOKENS

All Tokens Delete

Token

Token

Token

Token

Token

Token

Token

Token

Token

Token

Token Details

Use Token

Token Name Token

Access Token

eyJhbGciOiJSUzI1NiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICI3X3M5SXU3
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G9jYWxob3N0OjcwNzAvcmVhbG1zL2Vjb21tZXJjZS12aXJ0dWFsLX
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QtNDhlNy05MDllLTg5MzcxNDQ3ZmUxNSIsInR5cCI6IkJlYXJlciIsIm
F6cCI6ImVjb21tZXJjZS12aXJ0dWFsLWNsaWVudCIsInNpZCI6IjE1N
2NlOWQ2LWIzNTAtNDUwOS1hODlmLTIwNDcyMWMxNDU0NiIsIm
FjciI6IjAiLCJhbGxvd2VkLW9yaWdpbnMiOlsiaHR0cDovL2xvY2FsaG
9zdDo4MDgwIl0sInJlYWxtX2FjY2VzcyI6eyJyb2xlcyI6WyJkZWZhd
Wx0LXJvbGVzLWVjb21tZXJjZS12aXJ0dWFsLXJlYWxtIiwib2ZmbGI
uZV9hY2Nlc3MiLCJ1bWFfYXV0aG9yaXphdGlvbiJdfSwicmVzb3Vy

Client Secret

Client Secret

http://localhost:8080/api/v1/payment

Save

POST http://localhost:8080/api/v1/payment

Send

Params Authorization Headers (10) Body Pre-request Script Tests Settings

Cookies

Type OAuth 2.0

The authorization data will be automatically generated when you send the request. Learn more about authorization

Add authorization data to Request Headers

Client Authentication Send as Basic Auth header

Clear cookies

Get New Access Token

Store your secrets with end-to-end encryption locally using Vault.

Store in Vault

Body Cookies Headers (14) Test Results

Status: 200 OK Time: 685 ms Size: 874 B Save Response

Pretty Raw Preview Visualize JSON

"paymentLink": "https://checkout.stripe.com/c/pay/
cs_test_a1myzTu9gqGeixCI5Pq8zZcBLIn7ki2Fomcy5EJLYnnitCz223iGVlsXSU#fidnandhYHdWcXxpYCc%2FJ2FgY2RwaXEnKSdkdWxOYHwnPyd1blpxYHZxWjA0VjMzbmY0MHVmdG98X2Nmd0xmVlJjSUNAcDNiMUxfaWtOYUhDf0ZgYklHM
HJcfGxPbTJgcTBoRnByUldOa11RVGQ3ZkA8aXBMdDU1bnw3XU9dfEdqTz1BNTVyVHJLbnZvZicpJ2N3amhWYHdzYHcnP3F3cGApJ2dkZm5id2pwa2FGamlqdyc%2FJyZjY2NjY2MnKSdpZHxqcHFRfHVgJz8ndmxzYmlgWmxxYGgnKSdga2RnaWBVa
WRmYG1qaWFgd3YnP3F3cGB4JSUl"

mailtrap.io/inbox

Sandboxes My Sandbox

Search...

Payment for the order number 1
to: <user@example.com> minutes ago

Payment for the order number 1

From: <ecommerce-virtual@example.com>
To: <user@example.com>

Show Headers

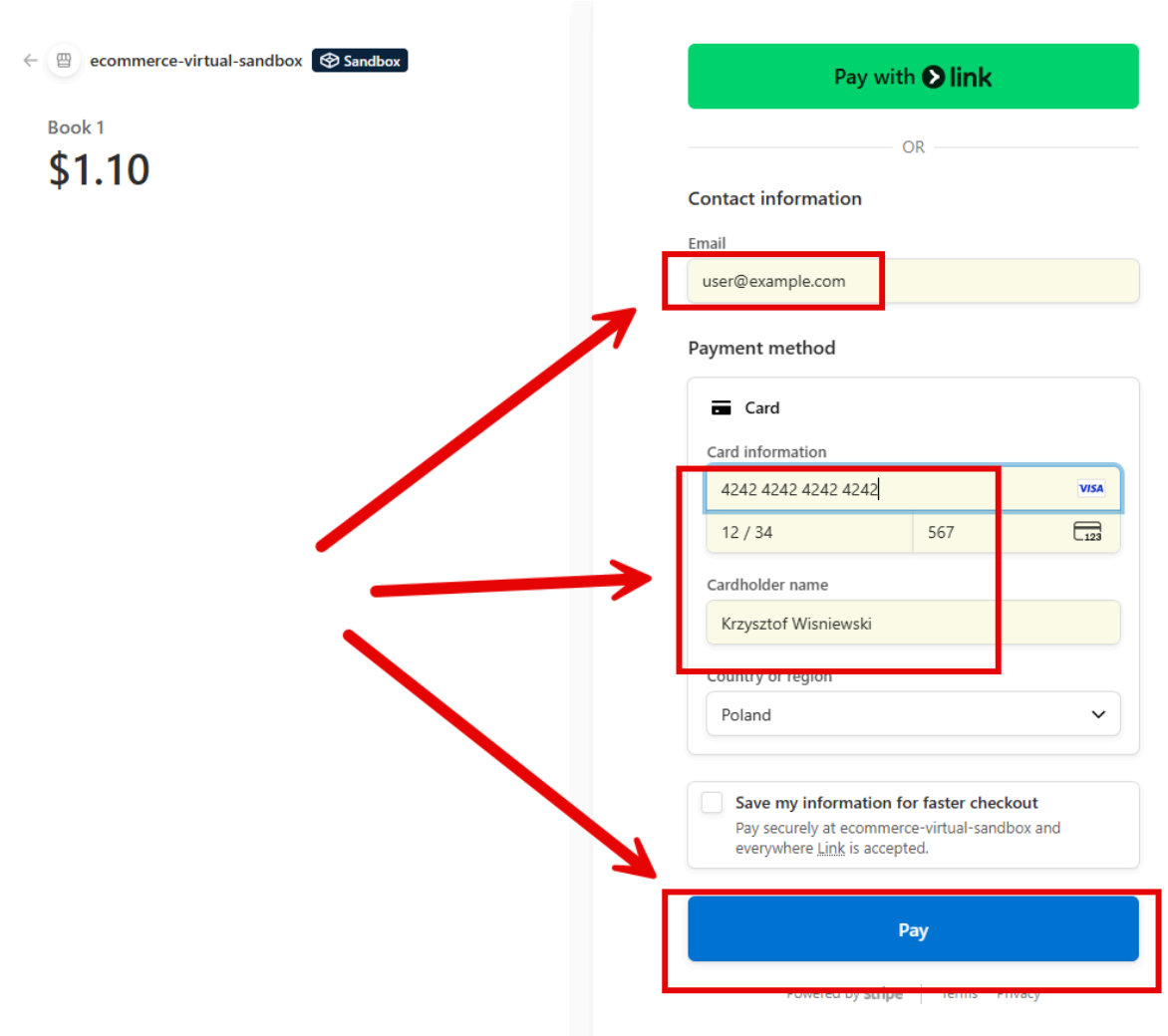
Looks good? Send to real users
Go live with Email API/SMTP - 4,000 production emails included

HTML HTML Source Text Raw Spam Analysis HTML Check 8 Tech Info

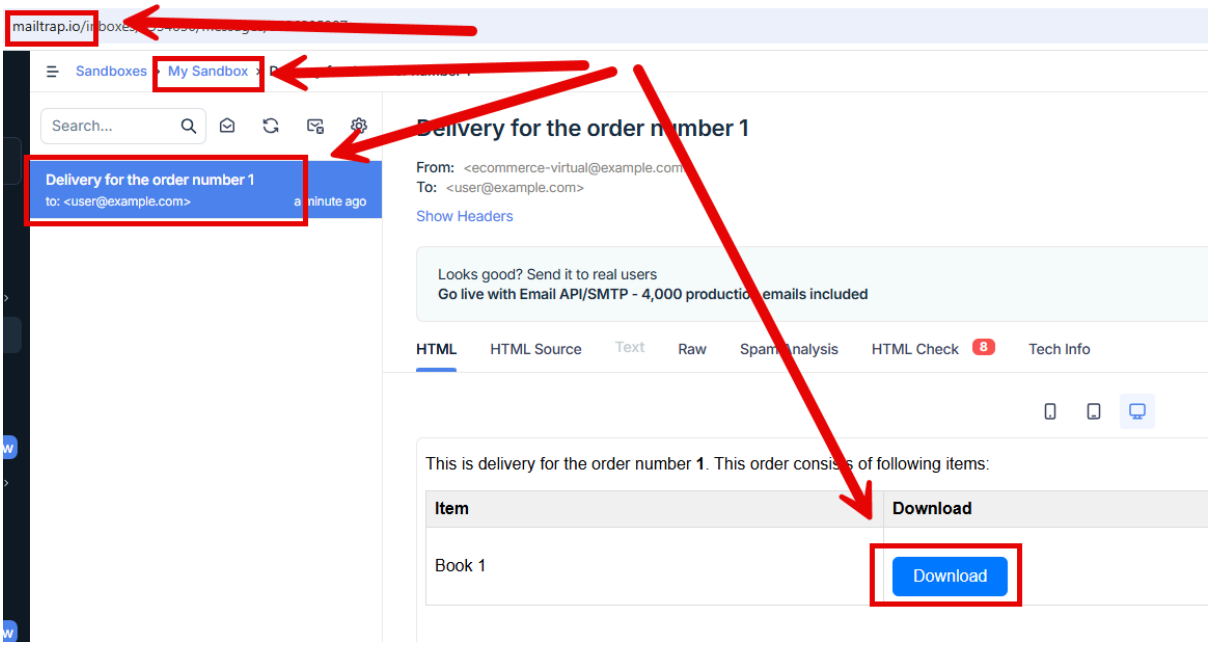
This is payment for the order number **1**. This order consists of following items:

Item	Price	Currency
Book 1	1.1	USD
Total	**1,1**	**USD**

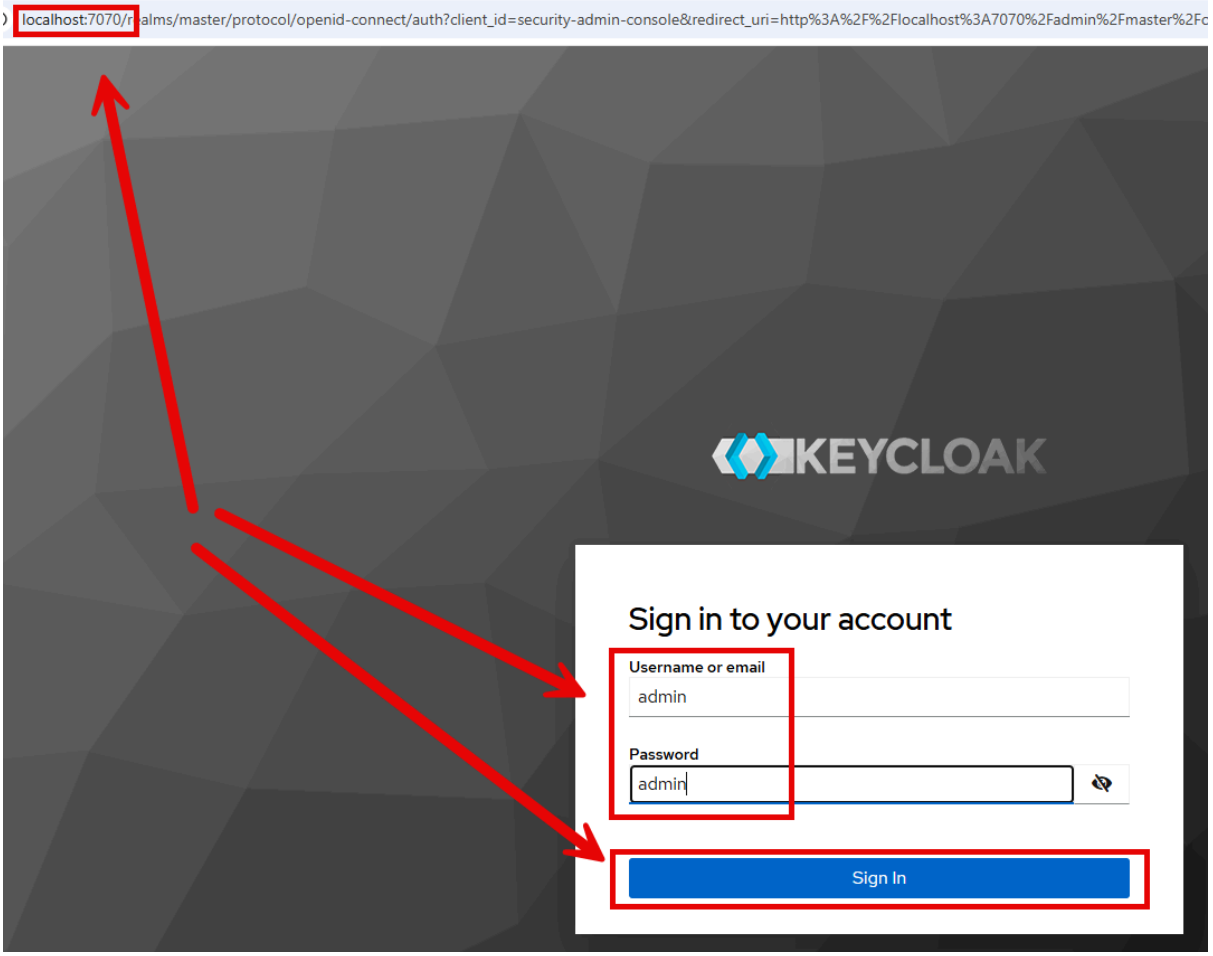
Payment

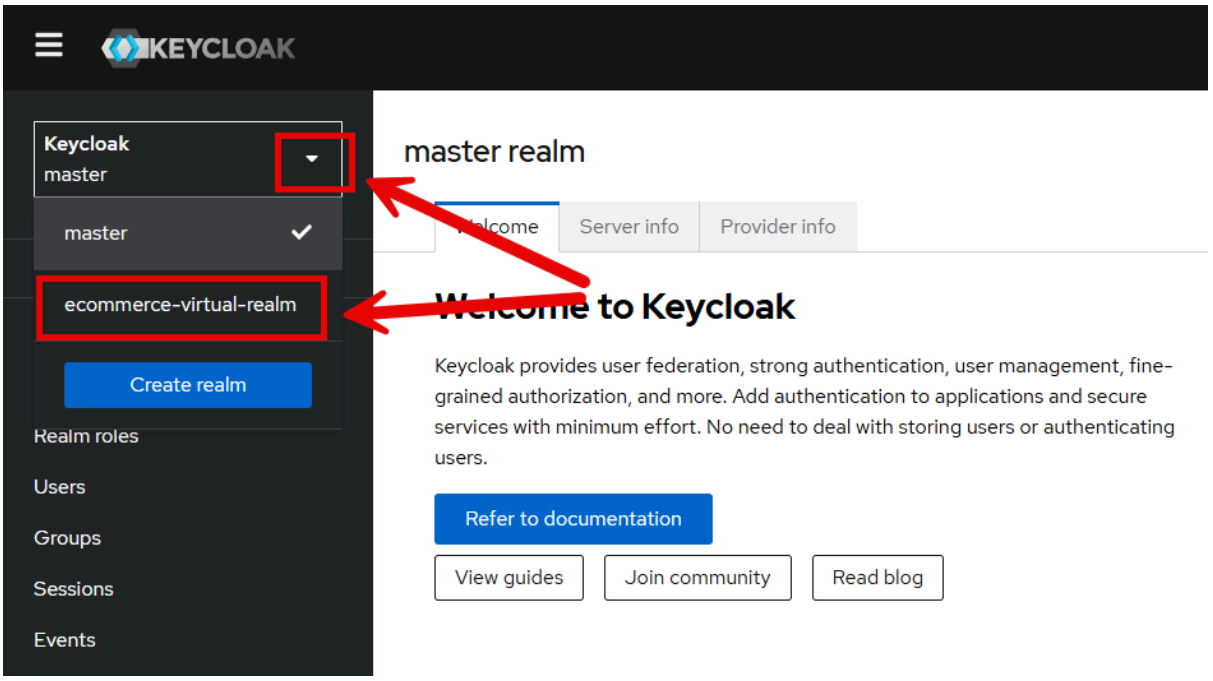

ecommerce-virtual-sandbox
Sandbox
Book 1
$1.10
Pay with link
OR
Contact information
Email
user@example.com
Payment method
Card
Card information
4242 4242 4242 4242
12 / 34
567
Cardholder name
Krzysztof Wisniewski
Country or region
Poland
Save my information for faster checkout
Pay securely at ecommerce-virtual-sandbox and everywhere Link is accepted.
Pay

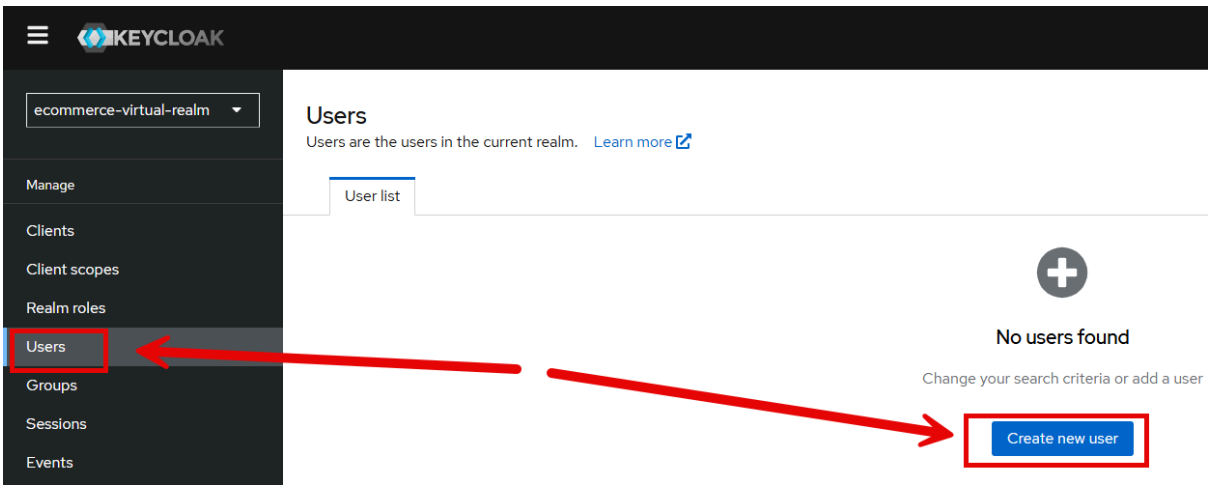
localhost:8080/api/v1/delivery?orderId=1&token=5141259712
Pretty-print
{"message":"Delivery successful for order 1"}

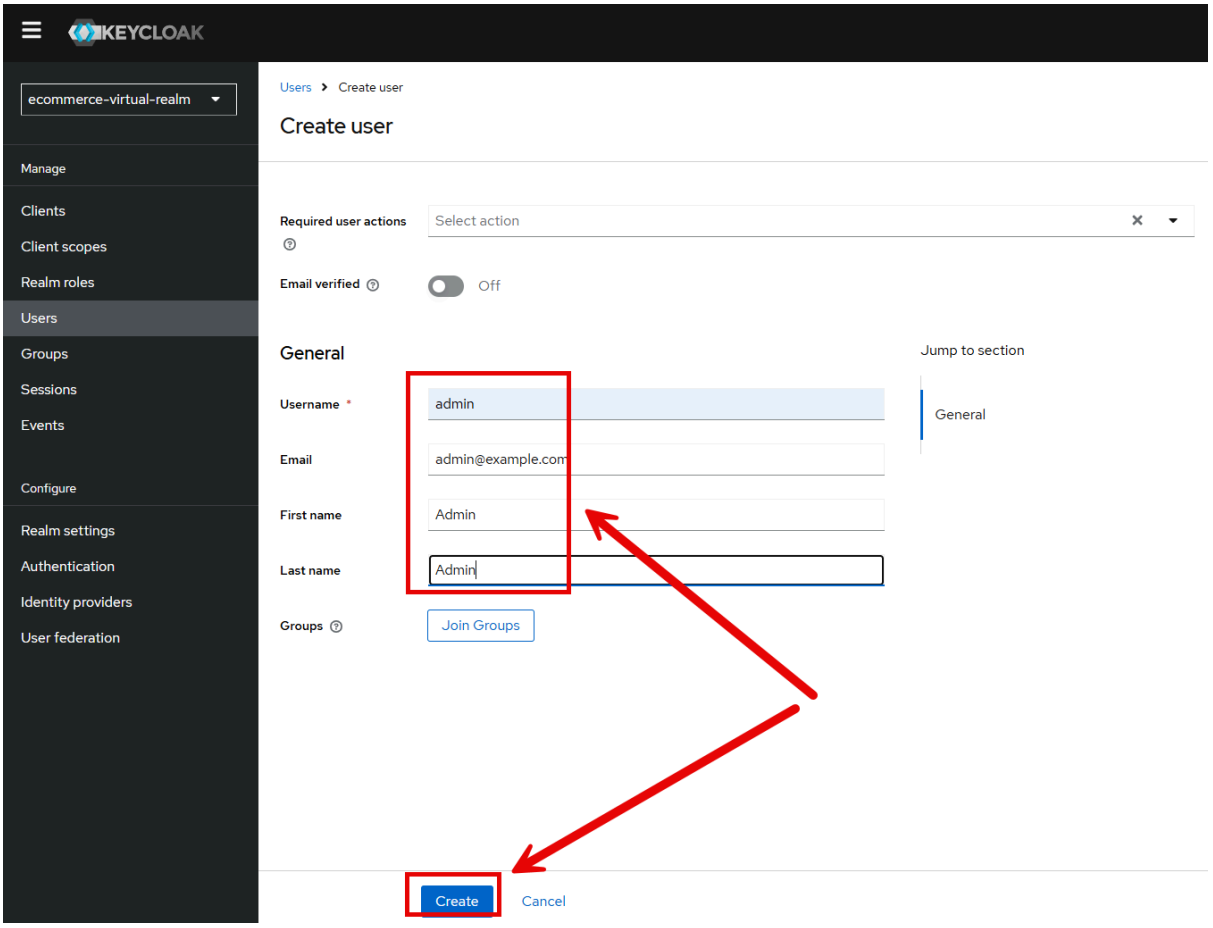

mailtrap.io/inboxes
Sandboxes
My Sandbox
Search...
Delivery for the order number 1
to: <user@example.com>
a minute ago
Delivery for the order number 1
From: <ecommerce-virtual@example.com>
To: <user@example.com>
Show Headers
Looks good? Send it to real users
Go live with Email API/SMTP - 4,000 production emails included
HTML
HTML Source
Text
Raw
Spam Analysis
HTML Check
8
Tech Info
This is delivery for the order number 1. This order consists of following items:
Item
Download
Book 1
Download

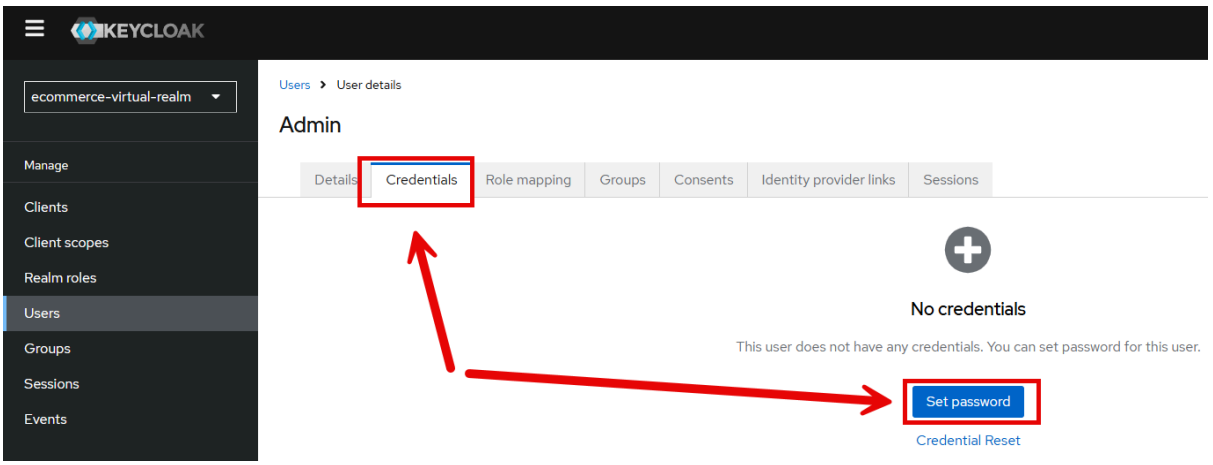
EXAMPLE SWAGGER 3

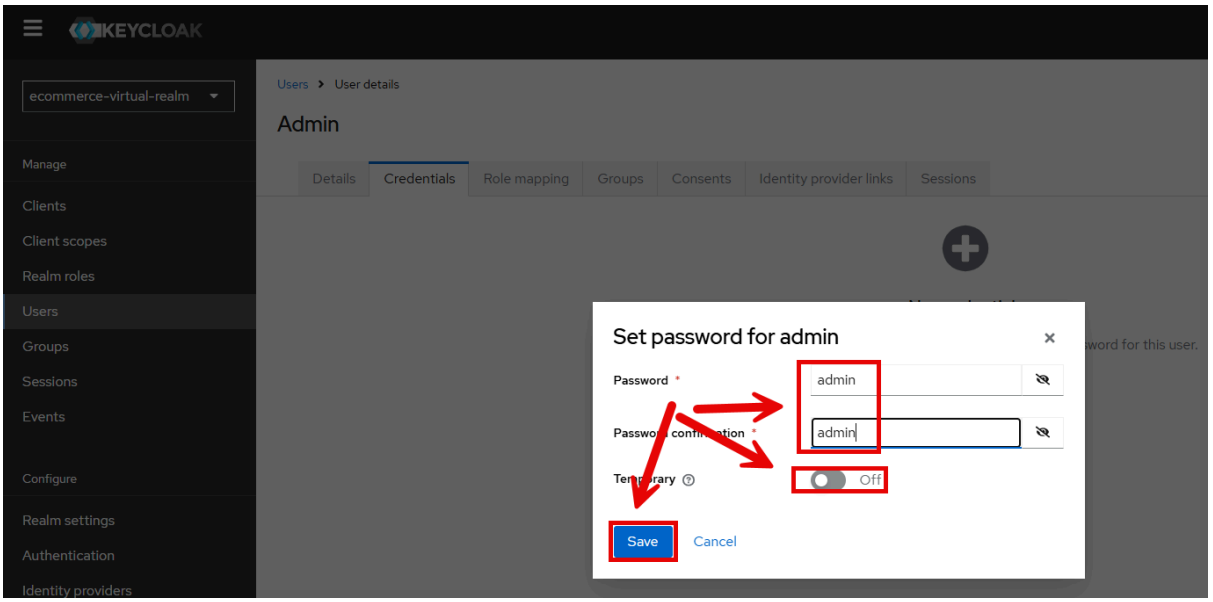


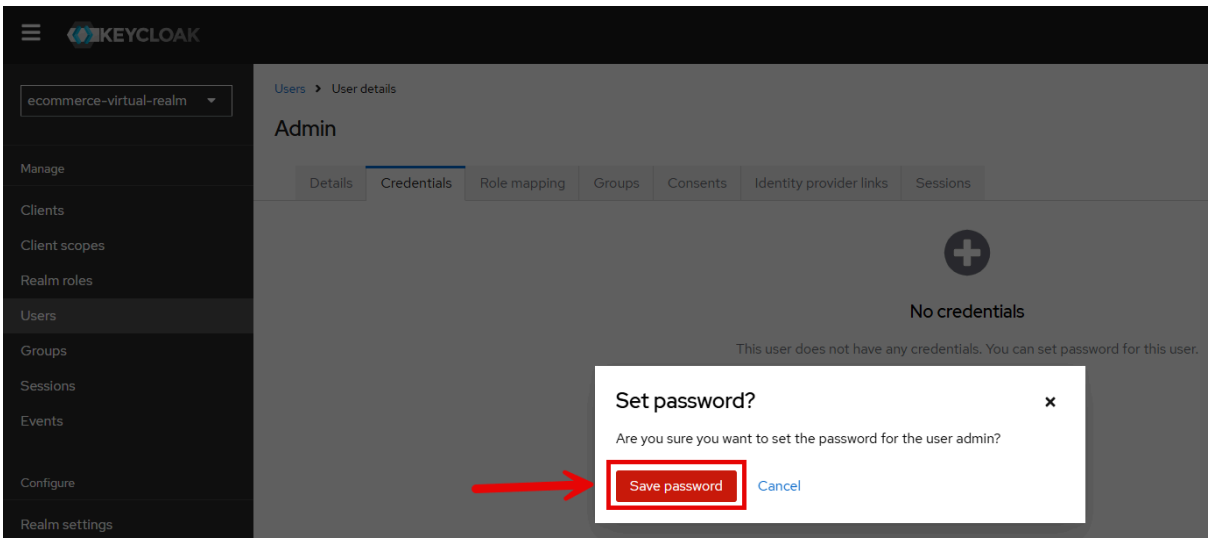


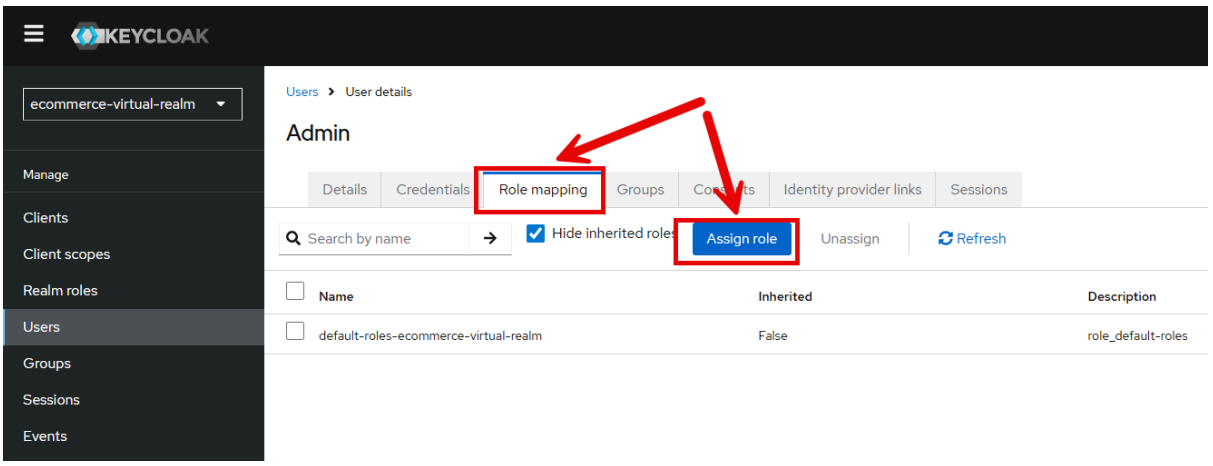

KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users
Users are the users in the current realm.
Learn more
User list
No users found
Change your search criteria or add a user
Create new user

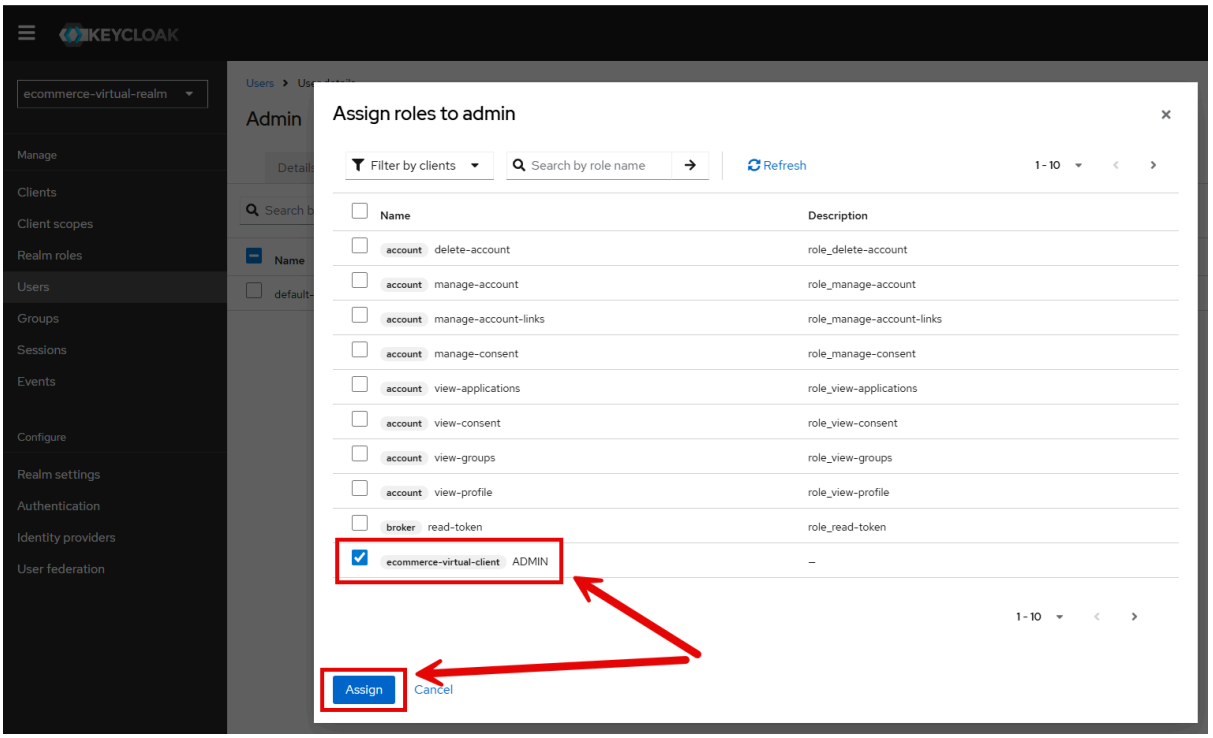

KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
User federation
Users > Create user
Create user
Required user actions
Select action
Email verified
Off
General
Jump to section
General
Username *
admin
Email
admin@example.com
First name
Admin
Last name
Admin
Groups
Join Groups
Create
Cancel

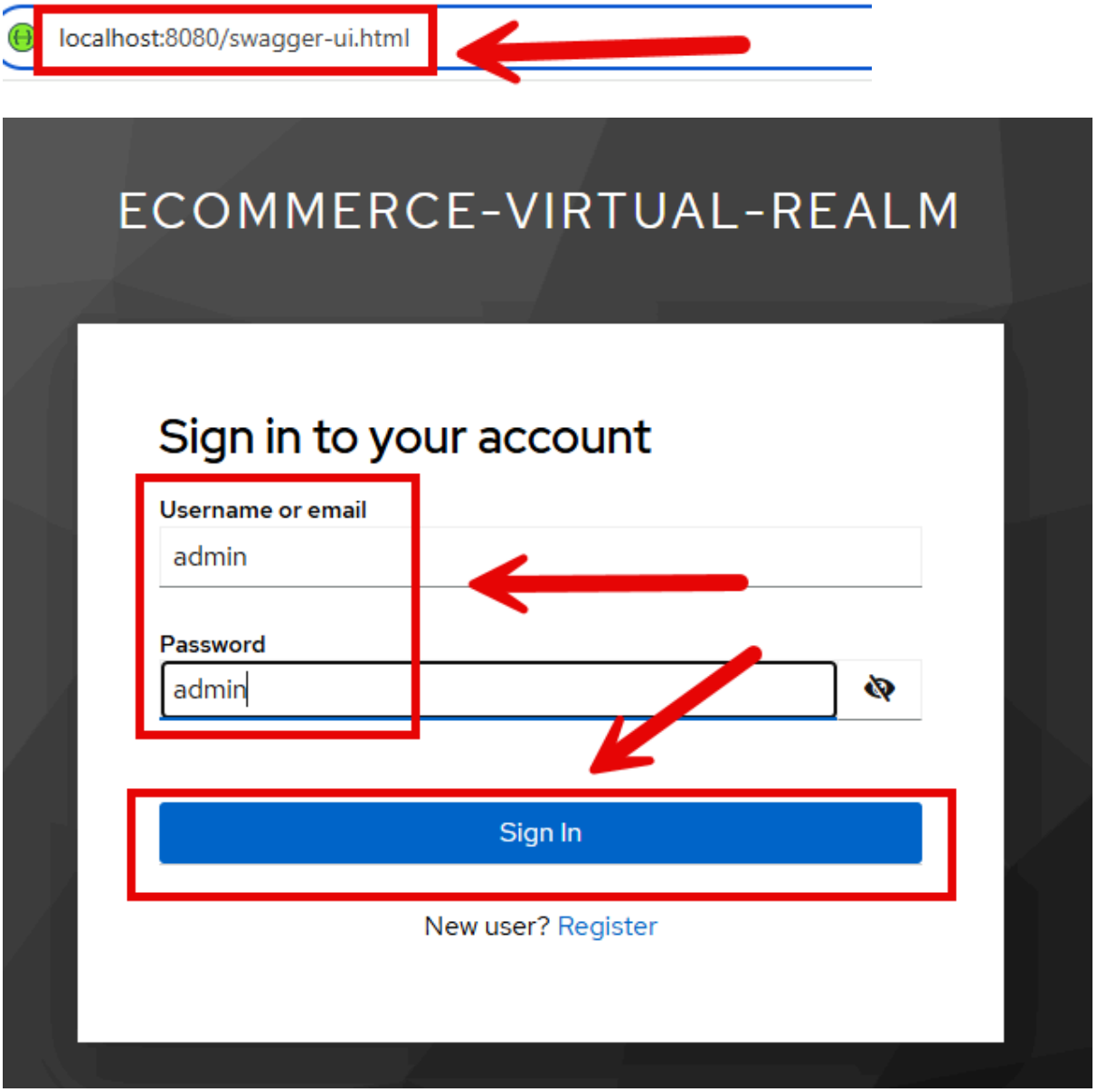

KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
No credentials
This user does not have any credentials. You can set password for this user.
Set password
Credential Reset


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
Set password for admin
Password *
admin
admin
Off
Save
Cancel


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Consents
Identity provider links
Sessions
No credentials
This user does not have any credentials. You can set password for this user.
Set password?
Are you sure you want to set the password for the user admin?
Save password
Cancel


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Users > User details
Admin
Details
Credentials
Role mapping
Groups
Identity provider links
Sessions
Search by name
Hide inherited roles
Assign role
Unassign
Refresh
Name
Inherited
Description
default-roles-ecommerce-virtual-realm
False
role_default-roles


KEYCLOAK
ecommerce-virtual-realm
Manage
Clients
Client scopes
Realm roles
Users
Groups
Sessions
Events
Configure
Realm settings
Authentication
Identity providers
User federation
Admin
Assign roles to admin
Filter by clients
Search by role name
Refresh
1 - 10
Name
Description
account delete-account
role_delete-account
account manage-account
role_manage-account
account manage-account-links
role_manage-account-links
account manage-consent
role_manage-consent
account view-applications
role_view-applications
account view-consent
role_view-consent
account view-groups
role_view-groups
account view-profile
role_view-profile
broker read-token
role_read-token
ecommerce-virtual-client ADMIN
–
Assign
Cancel


localhost:8080/swagger-ui.html
ECOMMERCE-VIRTUAL-REALM
Sign in to your account
Username or email
admin
Password
admin
Sign In
New user? Register

Swagger

Select a definition: Product Service

Product Service API

v0.0.1 OAS 3.1

/aggregate/product-service/v3/api-docs

This is the REST API for Product Service

Apache 2.0

You can refer to the Product Service Documentation

Servers

http://localhost:8081 - Generated server url

Products

REST API for Products

GET /api/v1/product Read Products

PUT /api/v1/product Update existing Product

POST /api/v1/product Create Product

Create new Product.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema


```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```



```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```


Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8081/api/v1/product' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}'
```

Request URL

http://localhost:8081/api/v1/product

Server response

Code Details

200

Response body


```
{
  "id": "1",
  "name": "Book 1",
  "description": "Description of Book 1.",
  "imageUrl": "https://upload.wikimedia.org/wikipedia/commons/8/8e/Alice%27s_Adventures_in_Wonderland_-_Carroll%2C_Robinson_-_S005_-_Alice%27s_Adventures_in_Wonderland.png",
  "locationUrl": "https://www.adobe.com/be_en/active-use/pdf/Alice_in_Wonderland.pdf",
  "price": 1.1
}
```


Download

Swagger

Select a definition: Order Service

Order Service API

v0.0.1 OAS 3.1

/aggregate/order-service/v3/api-docs

This is the REST API for Order Service

Apache 2.0

You can refer to the Order Service Documentation

Servers

http://localhost:8082 - Generated server url

Orders

REST API for Orders

GET /api/v1/order Read Orders

PUT /api/v1/order Update Order

POST /api/v1/order Create Order

Create new Order. Element 'id' is created automatically. Element 'products' is mandatory. Element 'email' is mandatory only in Swagger UI.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "id": 1,
  "email": "test@test.com",
  "products": "1,2,3",
  "orderStatus": "ORDER_INIT",
  "paymentLink": "http://payment.example",
  "token": "0123456789"
}
```

```
{
  "email": "user@example.com",
  "products": "1"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8082/api/v1/order' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "email": "user@example.com",
  "products": "1"
}'
```

Request URL

http://localhost:8082/api/v1/order

Server response

Code Details

200

Response body

```
{
  "id": 1,
  "email": "user@example.com",
  "products": "1",
  "creationDate": "2026-02-25T19:09:02.7204086",
  "modificationDate": "2026-02-25T19:09:02.7204086",
  "orderStatus": "ORDER_INIT",
  "paymentLink": null,
  "token": null
}
```

Download

Swagger

Select a definition: Payment Service

Payment Service API

v0.0.1 OAS 3.1

/aggregate/payment-service/v3/api-docs

This is the REST API for Payment Service

Apache 2.0

You can refer to the Payment Service Documentation

Servers

http://localhost:8083 - Generated server url

Payments

REST API for Payments

POST /api/v1/payment Create Payment

Create new Payment.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "orderId": 1
}
```

```
{
  "orderId": 1
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8083/api/v1/payment' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "orderId": 1
}'
```

Request URL

http://localhost:8083/api/v1/payment

Server response

Code Details

200

Response body

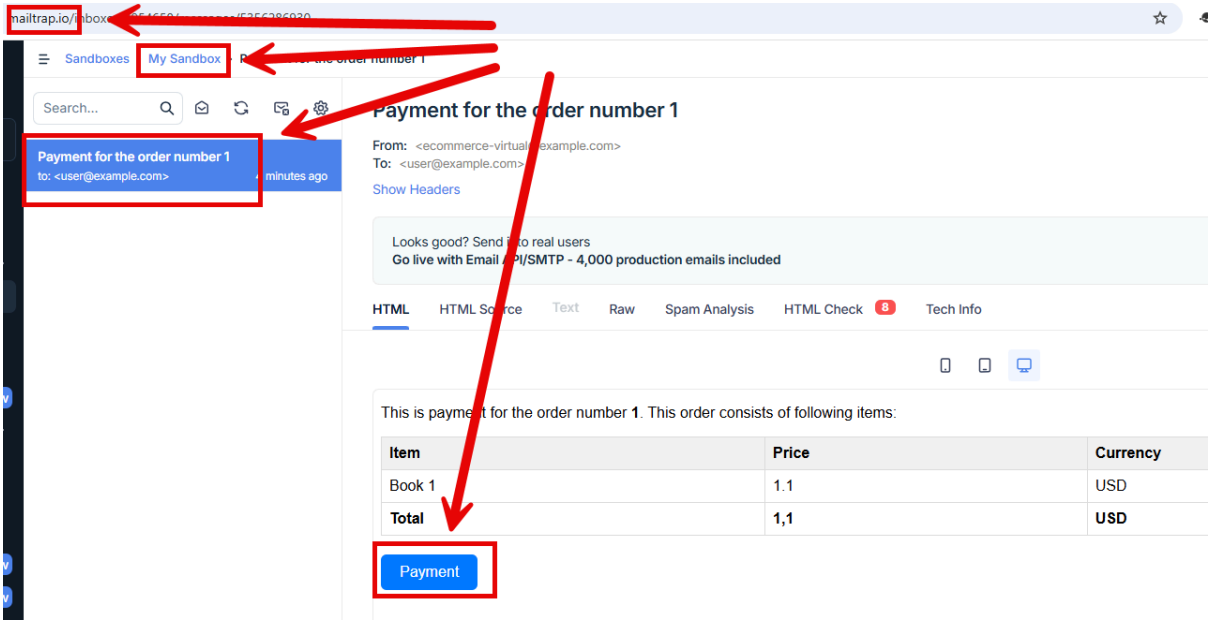

```
{
  "paymentLink": "https://checkout.stripe.com/c/pay/cs_test_a1vGdHOnbofSbsWcZvBhWKKTuWQ52Lv2xIe7famzNzNK3YQ3V1mUKKdOHnHfidnandhYHdWcXxpYCc%2FJ2FgY2RwaXEnKSdkdWxOYHwnPyd1blpxYHZxWjA0VjMzbmY0MHVmd
G9BX2Nmd0xmV1JjSUNAcDNiMUxfaWtOYUhDf0ZgYklHMHJcfGxPbTJgcTBoRnByU1dOa11RVGQ3ZkA8aXBMdDU1bnw3XU9dfEdqTz1BNTVyVHJLbnZvZicpJ2N3amhWYHdzYHcnP3F3cGApJ2dkZm5id2pwa2FGamlqdyc%2FJyZjY2N
HVgJz8ndmxrYmlgWmxxYGgnKSdga2RnaWBVaWRmYG1qaWFgd3YnP3F3cGB4J5UI"
}
```

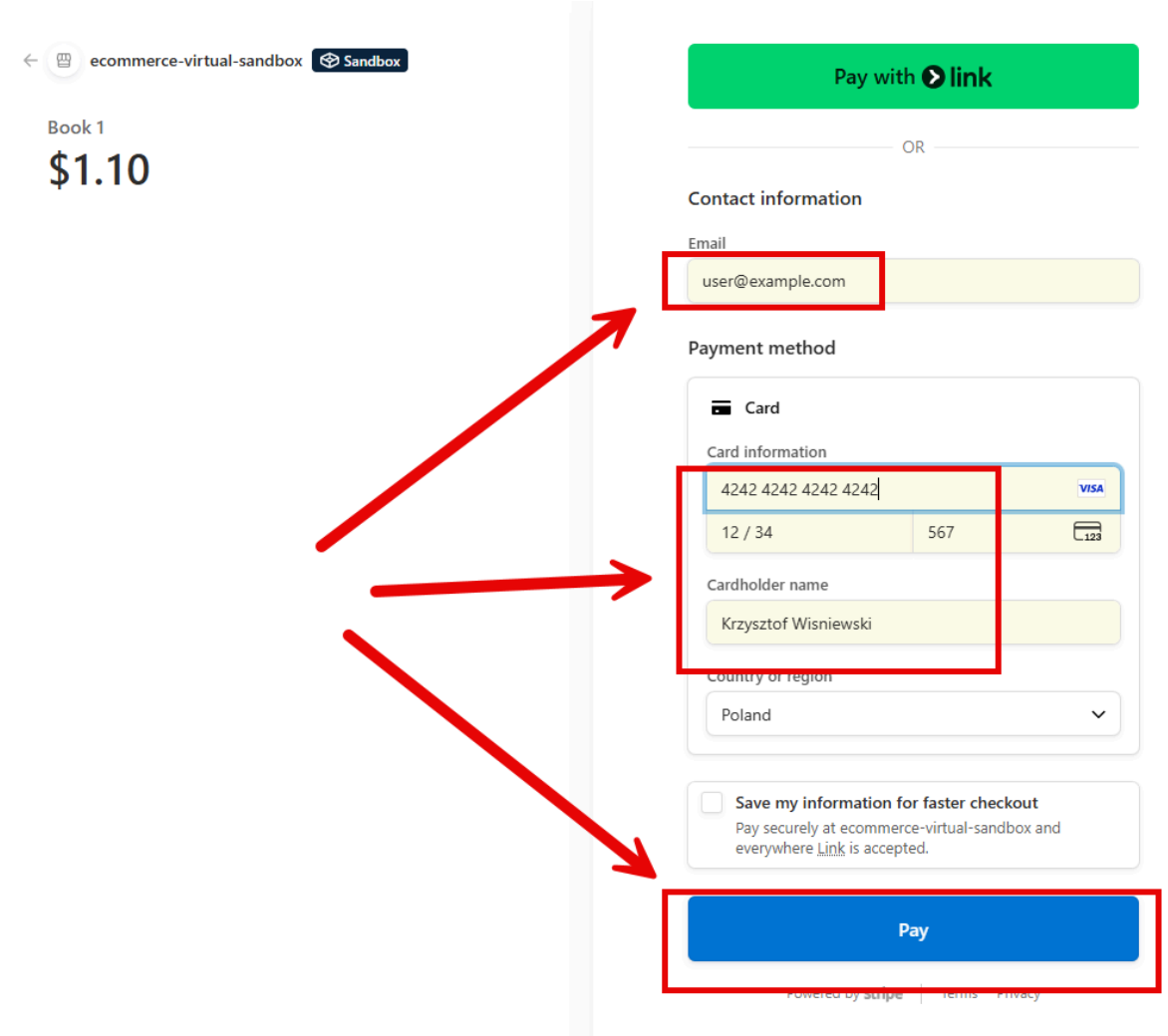

Download

Response headers

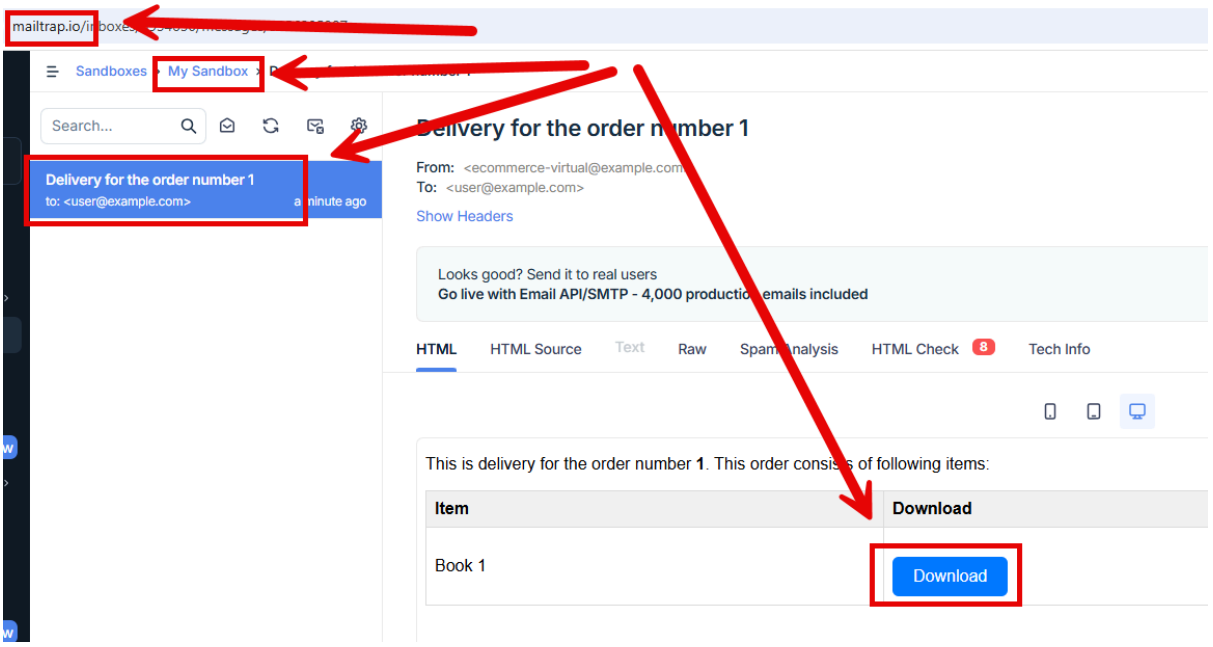
```
content-type: application/json
```

Responses

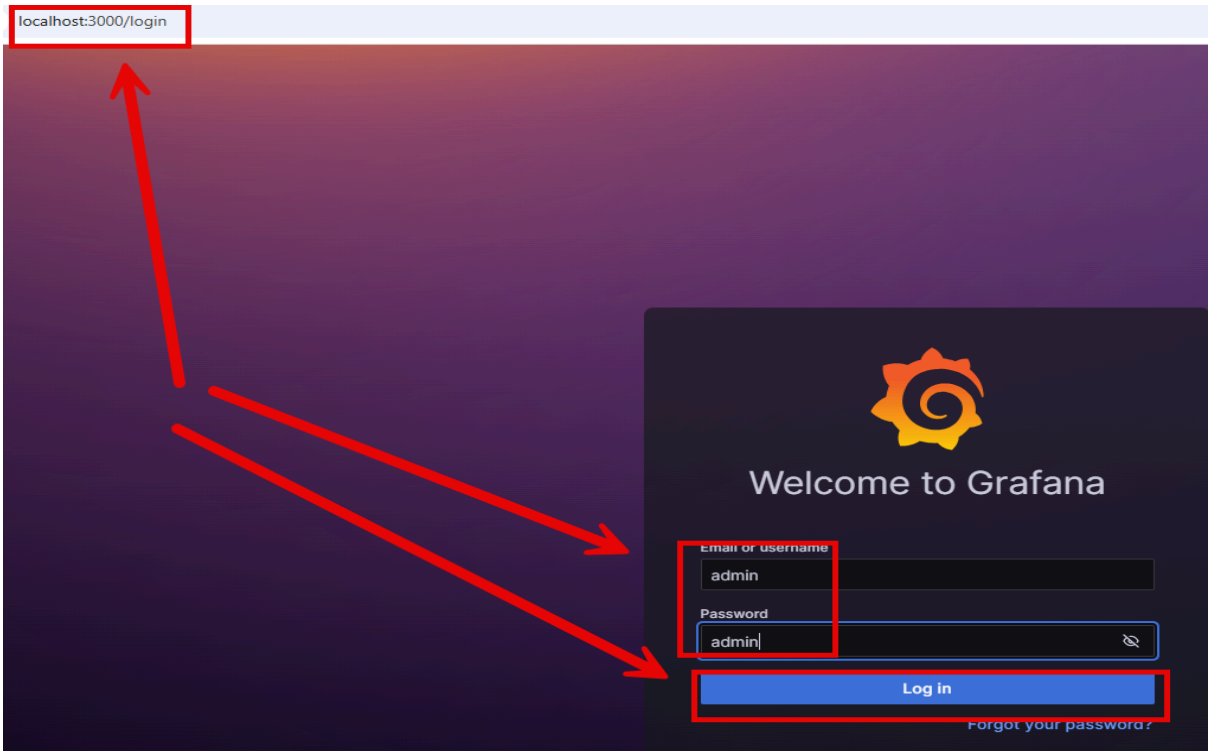

Sandboxes
My Sandbox
Search...
Payment for the order number 1
to: <user@example.com>
minutes ago
Payment for the order number 1
From: <ecommerce-virtual@example.com>
To: <user@example.com>
Show Headers
Looks good? Send to real users
Go live with Email API/SMTP - 4,000 production emails included
HTML
HTML Source
Text
Raw
Spam Analysis
HTML Check
8
Tech Info
This is payment for the order number 1. This order consists of following items:
Item
Price
Currency
Book 1
1.1
USD
Total
1,1
USD
Payment

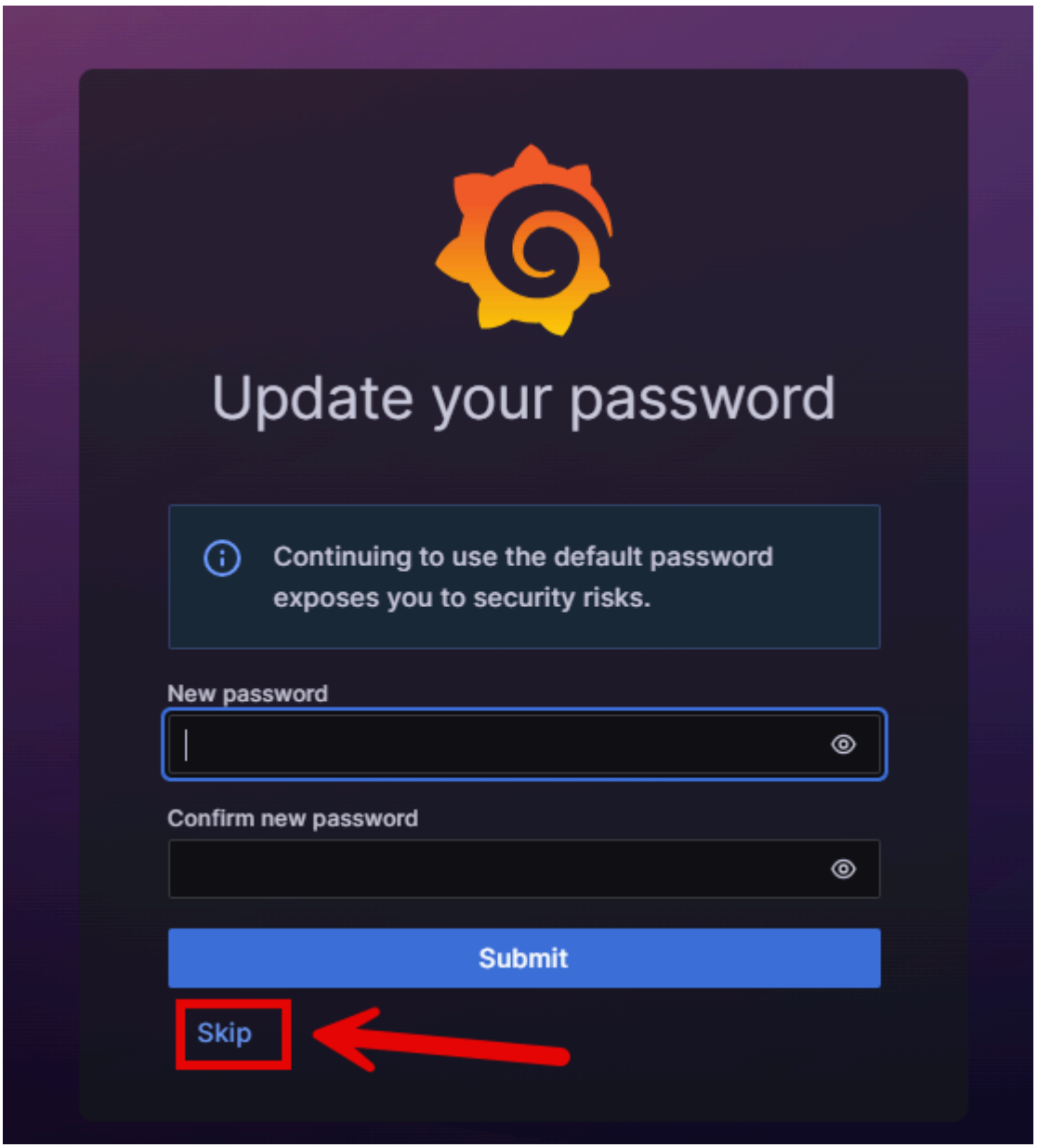

ecommerce-virtual-sandbox
Sandbox
Book 1
$1.10
Pay with link
OR
Contact information
Email
user@example.com
Payment method
Card
Card information
4242 4242 4242 4242
12 / 34
567
Cardholder name
Krzysztof Wisniewski
Country or region
Poland
Save my information for faster checkout
Pay securely at ecommerce-virtual-sandbox and everywhere Link is accepted.
Pay

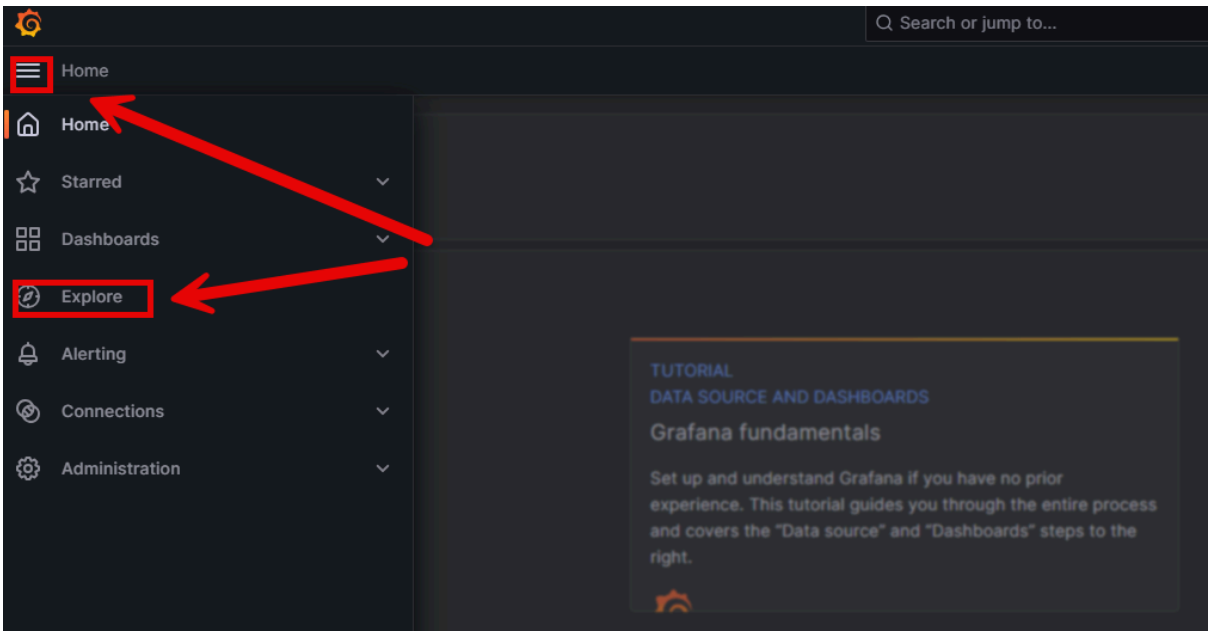
localhost:8080/api/v1/delivery?orderId=1&token=5141259712
Pretty-print
{"message":"Delivery successful for order 1"}

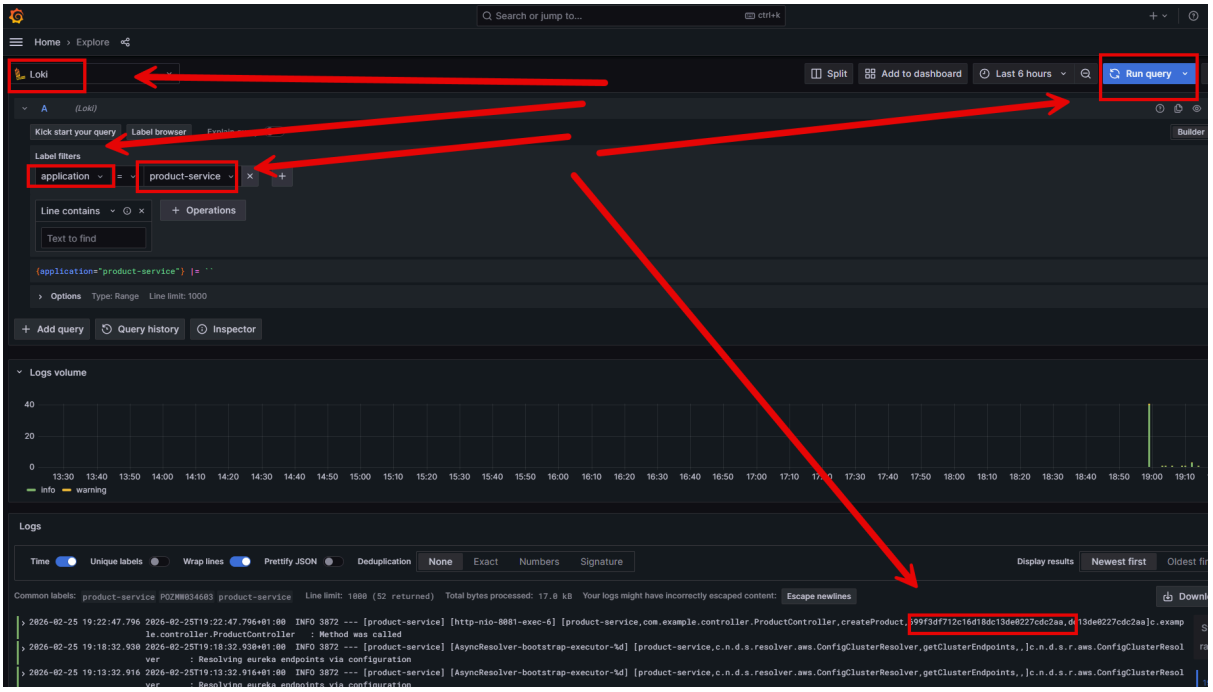

mailtrap.io/inboxes
Sandboxes
My Sandbox
Search...
Delivery for the order number 1
to: <user@example.com>
a minute ago
Delivery for the order number 1
From: <ecommerce-virtual@example.com>
To: <user@example.com>
Show Headers
Looks good? Send it to real users
Go live with Email API/SMTP - 4,000 production emails included
HTML
HTML Source
Text
Raw
Spam Analysis
HTML Check
8
Tech Info
This is delivery for the order number 1. This order consists of following items:
Item
Download
Book 1
Download

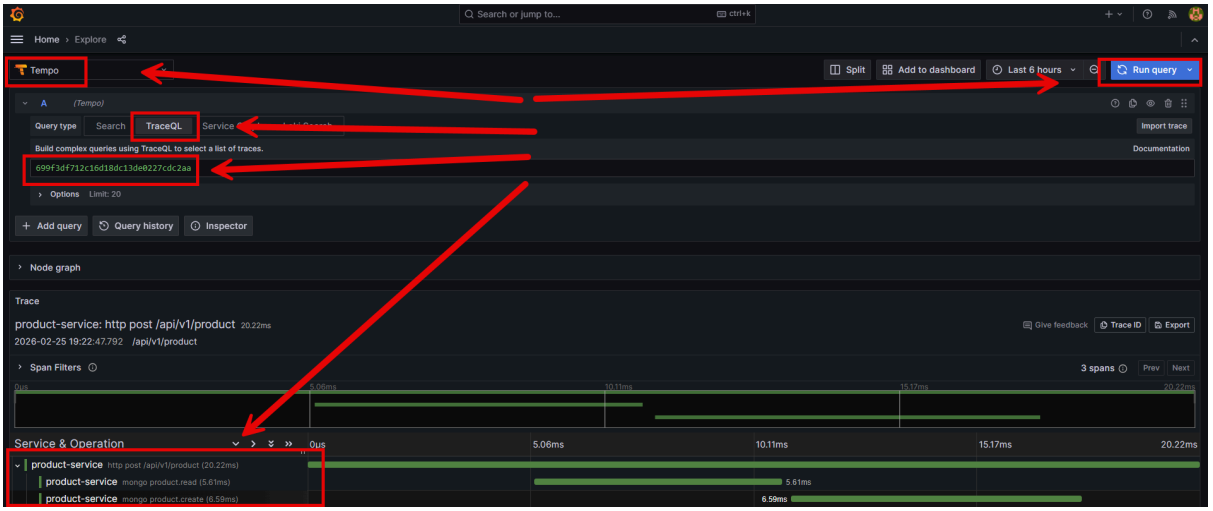
EXAMPLE GRAFANA



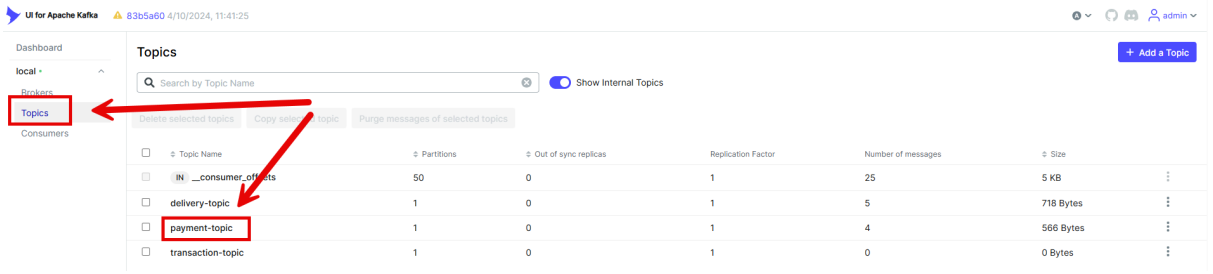

Update your password
Continuing to use the default password exposes you to security risks.
New password
Confirm new password
Submit
Skip

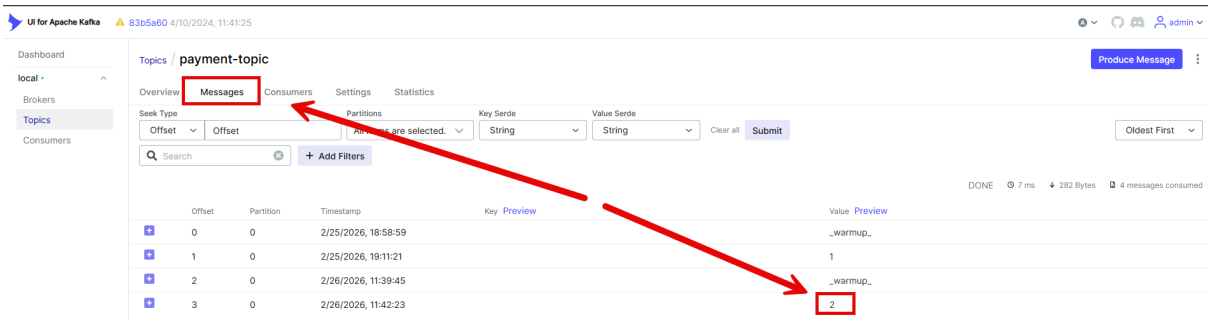

Search or jump to...
Home
Home
Starred
Dashboards
Explore
Alerting
Connections
Administration
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.


Search or jump to...
Home › Explore
Loki
Split
Add to dashboard
Last 6 hours
Run query
A (Loki)
Kick start your query
Label browser
Builder
Label filters
application
product-service
Line contains
Operations
Text to find
{application="product-service"} |= ``
Options Type: Range Line limit: 1000
Add query
Query history
Inspector
Logs volume
info
warning
Logs
Time
Unique labels
Wrap lines
Prettify JSON
Deduplication
None
Exact
Numbers
Signature
Display results
Newest first
Common labels: product-service POZMW034603 product-service
Line limit: 1000 (52 returned)
Total bytes processed: 17.0 kB
Your logs might have incorrectly escaped content:
Escape newlines


Search or jump to...
ctrl+k
Home
Explore
Tempo
Split
Add to dashboard
Last 6 hours
Run query
A
(Tempo)
Query type
Search
TraceQL
Service
Import trace
Build complex queries using TraceQL to select a list of traces.
Documentation
699f3df712c16d18dc13de0227cdc2aa
Options
Limit: 20
Add query
Query history
Inspector
Node graph
Trace
product-service: http post /api/v1/product 20.22ms
2026-02-25 19:22:47.792 /api/v1/product
Give feedback
Trace ID
Export
Span Filters
3 spans
Prev
Next
Service & Operation
0us
5.06ms
10.11ms
15.17ms
20.22ms
product-service http post /api/v1/product (20.22ms)
product-service mongo product.read (5.61ms)
5.61ms
product-service mongo product.create (6.59ms)
6.59ms

EXAMPLE KAFKA-UI





CONFIGURATION OF EXTERNAL SERVICES

Before running this project, you must configure the following external services:

- **Stripe** – External service for payments.
  Example configuration:
  https://github.com/wisniewskikr/chrisblog-it-java-springboot/tree/main/others/payment/springboot3-payment-stripe-api
- **Mailtrap** – External service for e-mails.
  Example configuration:
  https://github.com/wisniewskikr/chrisblog-it-java-springboot/tree/main/others/emails/springboot3-emails-text

CONFIGURATION OF ENVIRONMENT VARIABLES

Before running this project, you must configure the following environment variables:

- **MONGO_USERNAME** – Username of the NoSQL database for products in the e-commerce shop.
- **MONGO_PASSWORD** – Password of the NoSQL database for products in the e-commerce shop.
- **POSTGRES_USERNAME** – Username of the SQL database for orders in the e-commerce shop.
- **POSTGRES_PASSWORD** – Password of the SQL database for orders in the e-commerce shop.
- **STRIPE_SECRET** – Secret key for the external payment service (provided by Stripe).
- **MAILTRAP_USERNAME** – Username for the external e-mail service (provided by Mailtrap).
- **MAILTRAP_PASSWORD** – Password for the external e-mail service (provided by Mailtrap).
- **GRAFANA_USERNAME** – Username for the UI used for distributed logging, tracking, and performance monitoring.
- **GRAFANA_PASSWORD** – Password for the UI used for distributed logging, tracking, and performance monitoring.
- **KAFKAUI_USERNAME** – Username for the UI used for asynchronous communication.
- **KAFKAUI_PASSWORD** – Password for the UI used for asynchronous communication.
- **KEYCLOAK_USERNAME** – Username for the security tool used in the e-commerce shop.
- **KEYCLOAK_PASSWORD** – Password for the security tool used in the e-commerce shop.